

Does the Squeaky Wheel Get More Grease? The Direct and Indirect Effects of Citizen Participation on Environmental Governance in China[†]

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We conducted a nationwide field experiment in China to evaluate the direct and indirect impacts of assigning firms to public or private citizen appeals when they violate pollution standards. There are three main findings. First, public appeals to the regulator through social media substantially reduce violations and pollution emissions, while private appeals cause more modest environmental improvements. Second, public appeals appear to tilt regulators' focus away from facilitating economic growth and toward avoiding pollution-induced public unrest. Third, pollution reductions by treated firms are not offset by control firms, based on randomly varying the proportion of treated firms at the prefecture level. (JEL D22, L82, P28, P31, Q52, Q53, Q58)

Across the globe, 2.8 billion people breathe air that is considered hazardous by the World Health Organization, and 1.5 billion people contend with polluted water. A rapidly growing literature has documented severe consequences for health (Greenstone et al. 2015; Greenstone and Hanna 2014; Ebenstein et al. 2017), labor productivity (Graff-Zivin and Neidell 2012; Chang et al. 2016; Adhvaryu, Kala, and Nyshadham 2016), human capital accumulation (Isen, Rossin-Slater, and Walker 2017; Ebenstein and Greenstone 2022), and welfare (Kremer et al. 2011; Currie et al. 2015; Ito and Zhang 2020; Wang and Wang 2020). At the same time, most

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countries have strict standards and regulations on the books. It is apparent that the enforcement of regulations is failing in many parts of the world (Greenstone and Jack 2015; United Nations Environment Program 2019), but whether that is by design to facilitate economic growth (Greenstone, List, and Syverson 2012; He, Wang, and Zhang 2020) or due to genuine capacity issues (Duflo et al. 2013, 2018) is largely unknown.

An increasingly popular tool for reducing pollution is to encourage bottom-up participation in environmental governance. Programs that enable citizens, nongovernmental organizations, shareholders, and the media to participate in environmental enforcement date back at least to the 1980s when the United States introduced the Toxics Release Inventory (TRI), which required firms to publicly release their toxic emissions. In the subsequent 35 years, similar programs have been implemented in many countries, including Canada, China, India, and Indonesia.¹ Concurrently, many countries created official channels for the public to report pollution and violations of standards.² Yet there has been little rigorous evidence on whether and how citizens can leverage the government-released information to affect the enforcement of pollution standards and emissions at scale.

Additionally, we are unaware of any evidence on the indirect or general equilibrium impacts of bottom-up participation in environmental governance. In principle, the indirect impacts could reduce or even completely undo any direct impacts by reducing regulatory oversight of firms not subject to complaints. Alternatively, public participation could augment the direct effects by causing *all* firms to raise their expectations of the costs of violating pollution standards, leading to reductions in emissions by all firms through deterrence.

This paper evaluates the direct and indirect impacts of public participation in the enforcement of environmental standards in China, the world's largest polluter and manufacturer. A compelling feature of the study's context is that the Ministry of Ecology and Environment (MEE) maintains a Continuous Emissions Monitoring System (CEMS) that automatically collects hourly emissions data for 24,620 major polluting plants nationwide. These plants are responsible for more than 75 percent of China's total industrial emissions. These data are available in real time both to regulators and the public through provincial CEMS websites, so regulators in principle have all the information they need to detect and punish violations. Despite the widespread availability of these data, environmental compliance remains imperfect: in 2019, more than 33 percent of the CEMS firms violated pollution standards.

We conducted an eight-month nationwide field experiment that leveraged the CEMS data to file appeals against firms that violated standards. While appeals do not provide new information about the presence of violations, they inform regulators about which violations cause public dissatisfaction. We randomly assigned each CEMS firm to a control arm or one of several treatment arms that mirror the officially sanctioned ways that citizens and nongovernmental organizations already

¹ Besides the TRI, specific programs include the Greenhouse Gas Reporting Program (GHGRP) in the United States and Canada, the Maharashtra Star Rating Program (MSRP) in India, the Program for Pollution Control, Evaluation, and Rating (PROPER) in Indonesia, and the Continuous Emissions Monitoring System (CEMS) in China.

² For example, pollution hotlines and websites to file complaints against pollution violations can be found in countries like the United States, the United Kingdom, China, Canada, etc.

reveal their dissatisfaction about pollution to governments. The starting point for implementation was the daily determination of all firms that violated emissions standards the previous day. Specifically, when a treated firm committed a violation, citizen volunteers who cooperated with our experiment filed an appeal through one of two broadly defined channels: (i) *private appeals*, where the citizens complained to the regulator or the firm about the violation in ways that could not be observed by other members of the public, and (ii) *public appeals*, where the citizens complained about the violation on Weibo, a popular Chinese social media platform that is comparable to Twitter. These channels might imply different information to regulators about the nature of public dissatisfaction, the potential for collective action, and the costs associated with providing an unsatisfactory response. In total, the experiment intervened against nearly 3,000 pollution violations. This form of citizen appeal is a regular occurrence in China; for example, annually, there are roughly 300,000 appeals about industrial emissions registered with the Ministry of Environmental Protection by citizens and NGOs. Importantly, this experiment operated separately from the government's regulatory efforts, so our estimates are informative about the ways that the regulators respond to citizen complaints within existing regulatory practices.

Moreover, following Crépon et al. (2013), we cross-randomized treatment intensity at the prefecture level to assess whether indirect or general equilibrium consequences of appeals would offset their direct or partial equilibrium effects. In 60 percent of the prefectural cities, we assigned 95 percent of the CEMS firms to treatment arms, while in the remaining 40 percent of prefectural cities, we assigned 70 percent of the firms to treatments. Besides the direct effect on violators in the treatment group, the treatment could influence the behavior of firms assigned to treatment that do not violate standards and the entire control group. On the one hand, if firms believe regulatory effort is zero sum due to limited capacity, then the firms that are not targeted may increase emissions. On the other hand, if *all firms* in the 95 percent prefectures believe that regulatory oversight has generically increased, relative to firms in the 70 percent prefectures, then even firms that are not targeted will reduce their emissions. Whether one of these forces dominates is ultimately an empirical question.

There are three primary findings. First, public appeals on social media significantly reduced firms' subsequent violations and emissions, while private appeals to regulators and firms had a more modest impact. Specifically, firms in the public appeals treatment arm reduced violations by more than 60 percent, relative to the control group. Additionally, over the eight-month study period, the public appeals treatment caused sulfur dioxide (SO₂) emission concentrations to decline by 12.2 percent and reduced chemical oxygen demand (COD) emission concentrations by 3.7 percent relative to the control group. In contrast, even when using essentially the same content and wording as the public appeals, private appeals only caused approximately a 25 percent reduction in violations relative to the control group. The violations and emissions reductions were concentrated among the firms that frequently violated the standard prior to the experiment, especially those that significantly exceeded it.

Second, the available evidence suggests that the public appeals are so effective because they tilt local regulators' often competing goals away from facilitating

economic growth and toward avoiding pollution-induced public unrest. This conclusion is based on a qualitative examination of local regulators' career incentives and a series of quantitative tests. An especially important empirical finding comes from our randomization of the visibility of social media appeals about a violation by experimentally adding 10 additional "likes"/"shares" to the Weibo post. Appeals naturally receive an average of 0.66 likes/shares, so this is a substantial increase in visibility. This intervention caused regulators to become significantly more responsive: the probability of a reply to appeals increased by 40 percent, the length of written replies to appeals doubled, and the probability of an on-site investigation jumped by nearly 65 percent. In contrast, we fail to find evidence for several other potential mechanisms, including that the treatments caused firms to manipulate CEMS data.

Third, across a mix of outcomes, we find that the general equilibrium impacts do not offset the partial equilibrium effects; if anything, the interventions might have created positive spillover effects. For example, the relative probability of a violation was either unchanged or lower among the control and treatment firms in the high-intensity prefectures, although the reduced violation rates are not always statistically significant by conventional criteria. Overall, accounting for both general and partial equilibrium effects, we find that the ambient SO_2 concentrations declined by 3.5 percent in the 95 percent treatment prefectures relative to the 70 percent treatment prefectures (significant at 10 percent level).

This paper makes four contributions. First, this nationwide experiment allows us to offer unique insights into when and how citizen participation affects governance at scale. While citizen participation has long been promoted as the key to improving government accountability (Stiglitz 2002; Mansuri and Rao 2004; World Bank 2004), the existing literature has found mixed evidence on its effectiveness (Olken 2007; Banerjee and Duflo 2006; Björkman and Svensson 2009; Banerjee et al. 2012; Grossman, Platas, and Rodden 2018; Buntaine, Hunnicutt, and Komakech 2021; Buntaine, Zhang, and Hunnicutt 2021; Garbiras-Días and Montenegro 2022). These previous studies have addressed whether monitoring by citizens can improve governance by providing new, decentralized information about governmental performance. A distinctive feature of our study is that the key information on performance is already collected and disclosed by the government itself. This allows us to pin down how information about public demands changes the actions of governments (without confounding it with new information about the underlying governance outcome). Moreover, the unprecedented scale of our experiment permits us to vary the publicity, channel, and target of citizen appeals and thereby helps explain how different types of participation affect the enforcement of regulations by providing different signals of public discontent, at least in the Chinese context. Our paper is also the first to connect citizen participation, regulatory effort, and adjustments by firms, showing how participation can meaningfully decrease pollution.³

Second, this paper also relates to the burgeoning literature on social media's political and economic consequences. Existing papers have mostly focused on how

³Relatedly, there is also an emerging political science literature on authoritarian responsiveness, which studies the factors that determine whether an authoritarian government replies to citizens' requests (e.g., Chen, Pan, and Xu 2016; Distelhorst and Hou 2017; Anderson et al. 2019). We add to that literature by showing that the government not only replies to appeals and requests when faced with public pressure but also takes costly actions that result in actual improvements in government accountability.

social media shapes citizens' political attitudes and motivates political behavior (Bakshy, Messing, and Adamic 2015; Bursztyn et al. 2019; Allcott et al. 2020; Yanagizawa-Drott, Petrova, and Enikolopov 2021; Garbiras-Días and Montenegro 2022) and how it can foster collective action, both in China (Chen and Yang 2019; Qin, Strömberg, and Wu 2021) as well as in other authoritarian regimes globally (Steinert-Threlkeld 2017; Acemoglu, Hassan, and Tahoun 2018; Zhuravskaya, Petrova, and Enikolopov 2020; Enikolopov, Makarin, and Petrova 2020). Adding to this literature, our paper is the first to experimentally study how citizens can leverage social media to hold governments more accountable in policy enforcement, focusing on both partial equilibrium and general equilibrium consequences of using social media for bottom-up participation in governance.⁴

Third, we add to the literature on the political economy of environment. Existing work in this area has mostly focused on the strategic behavior of politicians in determining and implementing environmental policies (List and Sturm 2006; Kahn, Li, and Zhao 2015; Jia 2017; Greenstone et al. 2020) or the strategic interactions among local governments over environmental externalities (Burgess et al. 2012; Lipscomb and Mobarak 2016; He, Wang, and Zhang 2020; Wang and Wang 2020). In this paper, we provide experimental evidence on how pollution appeals by citizens hold local governments in China accountable in enforcing existing environmental standards. More generally, this paper also relates to the literature on the cost and benefit of different environmental policies (Henderson 1996; Greenstone 2002; Walker 2013; Ryan 2012; Kahn and Mansur 2013): our results demonstrate that mobilizing the public to engage in monitoring the performance of governments and firms might be a cost-effective way to improve compliance with existing environmental laws.⁵

Fourth, this paper bridges two strands of literature on the political economy of China's local governance model. The existing literature has pointed out that local governments in China have incentives to facilitate growth and provide support to the firms through both formal and informal institutions (Qian and Weingast 1997; Xu 2011; Bai, Hsieh, and Song 2020; Liu et al. 2022); and when dealing with the citizens, Chinese local governments have strong incentives to maintain local stability (Chen 2011; Lorentzen 2013; Campante, Chor, and Li 2019; Qin, Strömberg, and Wu 2021; Beraja et al. 2021; Wang and Yang 2021). Our paper connects these two lines of literature by documenting the interactions between the state-citizen relationship and the state-business relationship: when the public gets more involved in China's process of environmental governance, the regulatory relationship between government and polluting firms is reshaped; the result is increased governmental effort and lower pollution emissions by firms. Investigating the interactions between

⁴In a related paper, Mei and Wu (2022) use observational data to document how social media discussions of a vaccine-related scandal in China led to more transparent procurement of vaccines. Our paper echoes these findings in the context of enforcement of environmental regulation and our experimental design allows us to further compare the effectiveness of social media participation to other channels of public participation, as well as to investigate the associated general equilibrium effects.

⁵Relatedly, this paper clarifies the pathways by which transparency and information disclosure affect government and firm behavior. Increasing the amount of information disclosed to the public has become a common policy to improve regulatory and government performance (Gavazza and Lizzeri 2007; Mattozzi and Merlo 2007; Reinikka and Svensson 2011). Previous research focusing on disclosures about firms has focused on how transparency affects market capitalization (Konar and Cohen 1997; Bui and Mayer 2003). We provide evidence of a key alternate pathway that allows information disclosure to affect regulatory outcomes: by allowing the public to hold governments accountable for implementing policies effectively.

firms, citizens, and the state in a synthesized framework deepens our understanding of China's system for local governance.

The remainder of this paper proceeds as follows. Section I introduces the institutional background. Section II discusses our experiment and data. Section III presents the main empirical findings, while Section IV investigates the underlying mechanisms. Section V discusses the general equilibrium effects. Section VI concludes.

I. Institutional Background

This section describes the institutional background for the field experiment. Section IA details the encouragement and development of citizen participation in environmental governance in China. Section IB discusses China's Continuous Emissions Monitoring System. The section concludes with Section IC, summarizing our qualitative learnings about the dynamics between China's citizens, environmental regulators, and polluters and how we used them to develop hypotheses about which types and forms of citizen appeals might be effective in reducing environmental emissions.

A. Public Participation in China's Environmental Governance

China's unprecedented economic growth since the 1980s is accompanied by rapid industrialization and significant environmental degradation. The severe water and air pollution has led to increasing public discontent and social unrest across the country (Jing 2000; Steinhardt and Wu 2016), incentivizing the Chinese government to undertake significant regulatory changes aiming at improving its environmental quality. These measures include, for example, setting specific environmental performance targets (He, Wang, and Zhang 2020), launching nationwide pollution monitoring networks and providing real-time pollution information (Greenstone et al. 2022), and introducing pollution levies and taxes on large emitters (Gowrisankaran et al. 2020).

In addition to these "top-down" command-and-control-type approaches to environmental protection, the central government also explicitly encouraged "bottom-up" initiatives in the form of citizen participation. In 2006, the Ministry of Environmental Protection (MEP) issued the "Interim Measures for Public Participation in Environmental Impact Assessment," which emphasized the legal rights of citizens to get involved in making and implementing environmental policies. In the same year, the MEP established the "12369 Environmental Appeals Center," which hosted a national hotline (under the phone number 12369) that allowed citizens across the country to file environmental appeals about potential violations of pollution standards. Later, the platform expanded to include an official website. The MEP also instructed each prefectural city's environmental protection agency (EPA) to open an office to address citizen appeals.⁶

When a citizen makes an appeal via the 12369 platform, either by calling the hotline or leaving a message on the website, her appeal will be directed to the

⁶The online appeal platform can be accessed via <http://1.202.247.200/netreport/netreport/index>.

corresponding local EPA, which has legal responsibilities to investigate and issue fines to the polluter if a violation is confirmed. Between 2017 and 2019, the 12369 platform received a total of 1,860,149 appeals, 56 percent of which arrived by the hotline and the rest by the online platform. Appeals through the 12369 platform are visible to all levels of government.

In 2014, as part of China's grand "war on pollution," the central government introduced additional policies that further encouraged citizen participation in environmental protection, including the "Guiding Opinions on Promoting Public Participation in Environmental Protection" and the "Measures for Public Participation in Environmental Protection." In addition to reiterating the importance of the existing official channels for citizen participation in environmental protection, the central government required all the prefectural EPAs to set up official accounts on popular Chinese social media platforms, namely Weibo and WeChat, to make it easier for the public and local EPAs to communicate.

As of December 2017, all local EPAs in China's 338 prefectural cities operated official Weibo and WeChat accounts. In the past few years, an increasing number of citizens and NGOs have used these social media platforms to report environmental violations and express their discontent (Wu et al. 2021). For example, from 2014 to 2016, we identified 5,336 Weibo posts reporting alleged violations by CEMS firms, 1,563 of which were posted by NGOs and the rest by individual citizens. In 2018, a Jiangsu-based NGO called Public Environmental Concerned Center (PECC) alone filed 1,579 appeals on Weibo on pollution violations identified from the CEMS data.

B. China's Continuous Emissions Monitoring System

China operates the world's largest Continuous Emissions Monitoring System.⁷ The system was introduced in 2004 by the MEP, which consisted of the installation of automatic monitoring equipment, including apparatuses and flow (current) meters installed at discharge points, CCTVs covering all pollution prevention and control facilities, and data collection and transmission apparatuses. Each local EPA maintains a monitoring center that automatically collects data for each key pollutant from each installed meter in real time. Beginning in 2013, central policies required each provincial and prefectural EPA to publicize in real time the hourly emissions data of every monitored plant to the public. The data also include standards for different emission concentrations, allowing the public to check whether each plant violates the standards hour by hour.

The CEMS monitors the emission concentrations of both water pollutants (COD and $\text{NH}_3\text{-N}$) and air pollutants (SO_2 , PM, NO_x) for all the key polluters in China. If a firm's total emissions of any pollutant exceeded some threshold determined by the MEP in the past two years, it will be considered as a key polluter and thus be

⁷ Other countries use similar systems for various regulatory purposes. For example, the United States EPA and many state governments require firms to install CEMS equipment to demonstrate compliance with permitted emission levels (United States EPA 2021). In India, specific provinces have started to require firms to install continuous monitoring equipment to support emissions trading (Greenstone et al. 2020). Likewise, the European Union has made continuous emissions monitoring support operation of its Emissions Trading Scheme (EU 2021). Yet none of these schemes have approached the scale of the CEMS in China, where it is used as a systematic regulatory tool for all key industrial polluters.

continuously monitored by the CEMS. As of January 2020, the CEMS program monitored more than 24,620 plants, which accounted for more than 75 percent of China's industrial air and water pollutant emissions.

Our research hinges critically on the reliability of the CEMS data. The MEP exerts substantial effort to ensure the quality and authenticity of the CEMS data. First, the list of CEMS firms is publicized on the MEP website so that local governments cannot omit any CEMS firm from the publicized emission data. Second, the MEP has strict protocols for the installation and operation of the CEMS equipment: installation must be conducted by a third-party team designated by the MEP, and 24-hour CCTVs are installed near the monitoring equipment as a deterrent to the plant from interfering with the equipment. Third, the MEP uses various algorithms and technologies to detect abnormalities in the CEMS data and hosts monthly supervisory sessions to discuss any anomalies. Fourth, the MEP requires on-site inspections at least once a month to ensure the proper functioning of the automatic monitoring equipment. Additionally, the publicized CEMS data seem inconsistent with regular manipulation, as they include thousands of environmental violations across the country every month.

All in all, this set of safeguards leaves limited scope for the polluting plants and local EPAs to interfere with the CEMS. While these are all reasons to believe the data are highly reliable, Section IVB reports on tests for data manipulation and does not find evidence of it.

Despite the central government's efforts to collect and publicize high-quality data, environmental law delegates the enforcement of pollution standards to local regulators. Specifically, if the CEMS data indicate that a firm violates an emission standard, the local EPA is required to conduct an on-site investigation to verify the violation before punishing the firm. In addition, the issuance of substantial penalties, like large fines or shutdowns, involves two steps. First, the local EPA must issue a warning after detecting a violation through on-site inspection and give the firm one month to improve its practices. Then, another inspection will be conducted in a month, and the penalty can only be applied if the problem has not been addressed.⁸ It is thus not surprising that disclosing the CEMS data alone does not lead to perfect compliance, since large polluters can potentially defy or capture local regulators. Additionally, both the central and local governments have competing goals of maximizing production and controlling pollution. Maintaining some flexibility in applying emissions standards is likely to be desirable, especially when violations do not cause significant public discontent.

Figure 1 plots the percentage of emission violations using daily data from 2018 to 2020.⁹ In January 2018, around 1.5 percent of monitored stacks violated emission standards of air pollutants on any given day, and 0.6 percent violated emission standards of water pollutants. In the following three years, violation rates for both water and air pollutants declined steadily, which is consistent with national policy priorities. Nevertheless, around 0.9 percent of the monitored firms violated air pollution

⁸There are occasions where abnormal readings are recorded in the data, but they do not necessarily reflect actual emission violations. For example, this can happen when the monitor is malfunctioning, when the firm is turning on or turning off the equipment, or when the firm is switching between different production procedures. As a result, on-site verification is needed before firms can be punished for violations reflected in the CEMS data.

⁹Some CEMS firms can have multiple stacks being monitored, and they can be monitored for multiple pollutants.

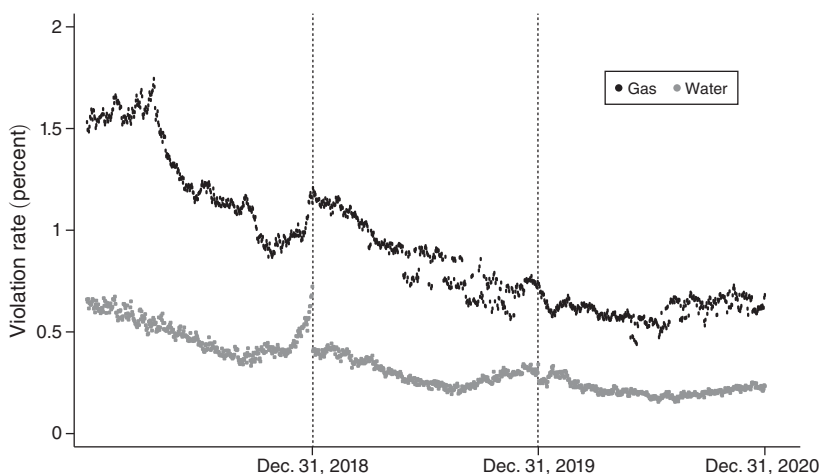


FIGURE 1. VIOLATION RATES OVER TIME

Notes: This figure plots the daily trends of waste gas and water emission violations in the CEMS data, between 2018 and 2021. The y-axis represents the ratio of CEMS firms' stacks that violated the gas/water emission standard on any given day.

standards, and 0.3 percent violated water pollution standards on any given day in the year before our experiment (2019). Bringing all CEMS firms into compliance with standards would significantly improve China's environmental quality: if violating firms reduced pollutant concentrations to just below the standards in 2019 (assuming no change in emission flows), SO₂ emissions would drop by 279,000 tons, a 7 percent reduction in aggregate industrial SO₂ emissions, and COD emission would drop by 31,000 tons, a 4 percent reduction in aggregate industrial COD effluents.

C. Interactions between Citizens, Regulators, and Polluters

To help design the treatments, we studied how citizens, regulators, and polluters interacted in the context of environmental governance, with special attention to the role of citizen appeals. These efforts include extensive conversations with environmental regulators, NGOs, polluting firms, and other relevant parties. Our visits with the directors and the staff assigned to respond to citizen appeals of two prefectural environmental bureaus in Hebei and Jiangsu provinces proved especially insightful.

First, we learned that each local environmental bureau has designated staff to handle citizen appeals filed. After an appeal is received, the staff member would route the appeal to the EPA's relevant office. For less severe violations, they might just call the polluting firm and collect relevant information. For more severe violations, a team of inspectors would conduct on-site inspections. The EPA, therefore, has significant discretion in levying penalties against violators.

Second, the regulators reported that they face two objectives that are often difficult to balance. On the one hand, political leaders, who could influence the career advancement of environmental regulators, emphasize the importance of economic growth. Local regulators are often given unofficial orders by local politicians that their regulations should not affect economic production and employment. This

typically means lax regulation for economically important local firms. On the other hand, local regulators also need to minimize the risk of social unrest caused by pollution. Both local politicians and regulators understand that social unrest and protests could be career ending. Therefore, the regulators must weigh the goal of preventing social unrest against growth pressures that lead to lax enforcement. Incentives pushed toward lax regulation when there is no public discontent and toward stringent regulation when violations cause public discontent.

Third, citizen appeals provide important signals of public discontent and the potential for collective actions. As a result, local regulators have incentives to monitor public sentiments and opinions online (especially on social media) and address individual complaints before they trigger wider public discontent. Responding to citizen appeals could also showcase their accountability, which might benefit their agencies in the long run. According to the internal rules of the two environmental bureaus we visited, all citizen appeals are expected to be addressed, unless the information provided in the appeals is inaccurate or could not be linked to a specific pollution source (e.g., there were frequent complaints of foul smells that were not connected to specific sources).

Finally, the regulators told us that the presumptive response to an appeal about a specific polluter is to send a team to the facility for an inspection. While regulators routinely conduct general-purpose inspections to check the paperwork and abatement facilities of firms as part of their normal operations, appeals often generate more task-specific inspections that involve testing pollution concentrations on-site. If emission violations get verified by on-site testing, the inspection teams could impose penalties and require the firms to make necessary corrections to their operations.

We accompanied the environmental inspection teams on a handful of inspections. An interesting observation from these visits was that almost all large polluting firms already had the capital equipment necessary to comply with the emission standards. However, firms were inclined to turn off certain equipment or skip abatement procedures occasionally due to the high marginal costs associated with the operation of such energy-intensive equipment. When this happened, the emission concentrations would elevate and eventually exceed the standards.

The fieldwork and qualitative evidence are instrumental in devising our experimental design. First, all forms of citizen appeals reveal information about public discontent, which can lead to more frequent environmental inspections and raise firms' expected costs of violating environmental standards. Second, public appeals can be more effective than private appeals because the information about both the violation and discontent is observable to the general public, which may lead to collective actions. Third, social media can be an effective and inexpensive way to improve government accountability and reduce emission violations. This is because social media appeals can be observed by the general public and have high potentials to trigger collective actions.

II. Experiment and Data

This section describes the field experiment and data. Section IIA discusses the experimental design, Section IIB provides details on the experiment's implementation, Section IIC discusses ethical considerations, and Section IID introduces the data and presents balance tests across the experimental arms.

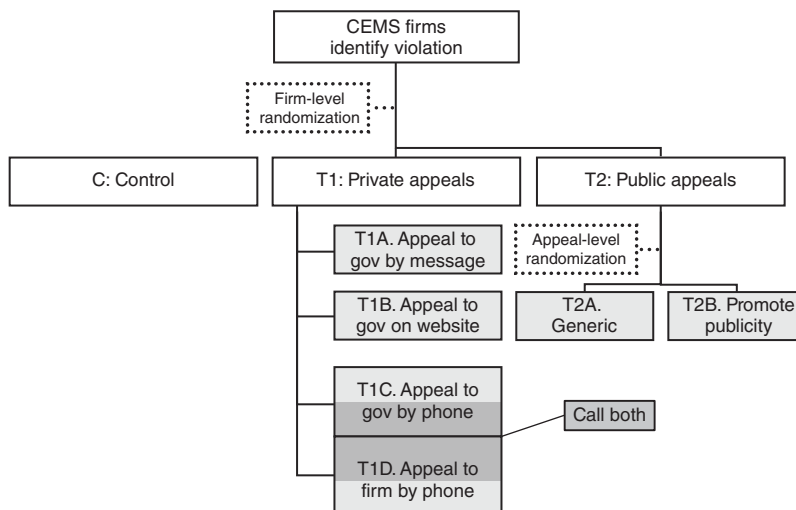


FIGURE 2. EXPERIMENTAL DESIGN

Note: This figure illustrates our experimental design, in which each CEMS firm is randomly assigned to one of seven different arms.

A. Experimental Design

The sample is comprised of the 24,620 polluting firms required to install CEMS by the central government by January 1, 2020. We randomly assigned these firms to several experimental arms designed to uncover the effects of private and public appeals, to shed light on the mechanisms that explain government and firm responses, and to learn whether there are general equilibrium impacts. The main outcomes of interest include each firm's daily violation status and daily average air and water pollution emissions concentrations as well as prefecture-level ambient SO₂ concentrations.

Experimental Arms: Public and Private Appeals.—Figure 2 graphically depicts how these firms were randomly assigned to three broad groups of experimental arms: the control group (C), the “private appeals” group (T1), and the “public appeals” group (T2). We use these experimental groups to identify the causal effects of different appeals relative to the status quo regulatory response and each other. Importantly, the treatments mirror existing and approved ways that citizens participate in environmental governance. Specifically, the three experimental arms and the proportion of firms assigned to them are

- **Control Group (C):** When the CEMS data indicated that the firm violated its emission standards, we *did not* intervene in any way. About 1/7 of the CEMS firms were assigned to this group.
- **Private Appeals Group (T1):** When the CEMS data indicated that the firm violated its emission standards, a volunteer filed a private appeal against that violation that was not observable to the public. About 5/7 of the CEMS firms were assigned to this group.

- **Public Appeals Group (T2):** When the CEMS data indicated that the firm violated its emission standards, a volunteer wrote a post on Weibo (a popular Chinese social media platform comparable to Twitter) and tagged (“@”) the official Weibo account of the corresponding local EPA. The appeal and any response were observable to the public. We assigned 1/7 of the CEMS firms to this group.

For each appeal arm, we prepared a detailed script for the citizen volunteers to follow. The core content of these scripts remained consistent across T1 and T2, while we randomly varied the exact wording in each appeal to avoid appearing repetitive to the regulator or firm. The volunteers who made appeals did not reveal information about their background or organizational affiliation, such that the appeals mirrored how the public normally participates. Samples of the appeal scripts are translated and listed in online Appendix B.

Mechanisms.—Within the groups of experimental treatments, we further randomized firms into specific treatment arms to investigate several potential mechanisms for the overall private (T1) and public (T2) treatment effects. Specifically, the T1 private appeals were delivered in several different ways following the MEE’s recommended channels for appeals about environmental issues:¹⁰

- **Private Appeals to Regulator via Direct Message on Social Media Group (T1A):** A citizen volunteer sent a *private* message to the corresponding local EPA’s official Weibo account, notifying them about the pollution violation and requesting that they investigate the issue.
- **Private Appeals to Regulator on Government Website Group (T1B):** A citizen volunteer filed a *private* appeal via the 12369 website to the corresponding local EPA, notifying the local EPA about the violation and requesting that they investigate the issue.
- **Private Appeals to Regulator through Government Hotline Group (T1C):** A citizen volunteer called the 12369 hotline to *privately* appeal to the corresponding local EPA. In the phone call, she notified the local EPA about the violation and requested that they investigate the issue.
- **Private Appeals to Firm through Phone Call (T1D):** A citizen volunteer called the violating firm to *privately* appeal the violation. In the phone call, she notified the firm about its violation and requested that they check the issue.¹¹

Furthermore, we cross-randomized T1C and T1D, such that half of the firms receiving T1C also simultaneously received T1D and vice versa. The sub-arms in T1 are

¹⁰Naturally, our interventions cannot exhaust all the possible channels through which private pollution appeals can be filed. Nevertheless, we believe that the subset of appeal channels that we choose are the most common types of private pollution appeals in China. They were also explicitly endorsed by the MEE itself in its guidelines for citizen participation in environmental governance.

¹¹The phone numbers we used to contact the firms were the official numbers listed on the updated business registration records, which are the same numbers that governments and other businesses would use to contact these firms.

randomized at the firm level. T1A and T1B each account for 1/7 of the CEMS firms, while T1C and T1D jointly account for 3/7 of the CEMS firms.

A comparison of the T1A–T1C treatment effects with the T1D treatment effect reveals whether private appeals to local governments are more effective than private appeals to firms. Further, a comparison of the T1A–T1C treatment effects provides an opportunity to assess whether the government is more responsive to appeals through a particular channel. For example, an appeal delivered through a social media direct message (T1A) might indicate a greater risk that the violation will draw widespread attention than an appeal delivered via an older technology like a phone call (T1C). The interaction term between T1C and T1D created by our cross-randomization provides a test of whether there are complementarities in privately appealing to regulators and to firms. Additionally, this interaction is indirectly informative about the nature of the T2 treatment effect since public appeals by their very nature involve informing both the regulator and the firm of the violation.

Within the T2 treatment group, we randomly assigned half of the violations to receive additional public attention by hiring a social media firm to increase the number of “likes” and “shares” for these Weibo posts at the *appeal* level (T2B). Ultimately, the average number of “likes” and “shares” were 10.56 for T2B, compared to 0.66 for T2A, which did not include any promotion. Additionally, online Appendix Figure A1 shows that the “likes” and “shares” in T2B amounted to 3–4 times the average level of engagement with pollution appeals posted by other NGOs or posts made by the local regulators. Further, 10.56 likes/shares is at the ninety-ninth and ninetieth percentiles of the distributions for posts from these organizations. Given these benchmarks, we believe that our “visibility promotion” treatment of adding an additional 10 likes/shares is a substantive boost to the visibility of the social media appeals and likely to draw the attention of the regulators who want to avoid instances of social unrest due to environmental concerns.

Thus, the comparison between these two arms provides a test of publicity or visibility about pollution violations, which might be important given the regulator’s objective to avoid collective actions and civil unrest. Among the T2B and T2A violations, we compare whether the Weibo appeal receives a response from the regulator, the response’s length measured in words, and whether the response includes proof of an on-site inspection or audit of the violator. These outcome variables are collected through our Weibo exchanges with the local regulators and are not available for the other arms.¹²

General Equilibrium.—We also experimentally investigate the indirect or general equilibrium consequences of the pollution appeals. Although the public and private appeals are applied at the firm level, they could also affect the ways that regulators enforced standards generally within their jurisdiction. On the one hand, regulatory

¹²It is worth noting that, even if we can collect these violation-level outcome variables for all violations, we still will not be able to draw any causal conclusion from a comparison across C, T1, and T2. This is because the three main arms were randomized at the *firm* level, and during the experiment, any violation after the first one will be endogenous to the treatment received by that firm. As a result, the types of firms that keep committing violations, as well as the frequencies and severities of these subsequent violations, could differ across C, T1, and T2 and thereby mechanically lead to differences in regulatory responses and efforts. We can compare these outcome variables across T2A and T2B because these two subtreatments were randomized at the *appeal* level.

resources may be fixed, and their application to violators in the treatment group may crowd out the regulatory resources that would otherwise have been allocated to control firms or treatment firms that previously did not violate standards, potentially leading to an increase in emissions and violations by them. On the other hand, the treatment might cause a positive spillover effect by (i) leading local governments to generically enforce environmental regulations because they interpret the increased appeals as an indication of the public's broad dissatisfaction with environmental quality and (ii) causing these firms to proactively reduce their emissions because they interpret the increased appeals and enforcement against violators as evidence of an increase in regulatory stringency. These broad forces work in opposite directions and whether one dominates is ultimately an empirical question.

To explore these possibilities, we cross-randomize treatment intensity across different regions. Specifically, 95 percent of the CEMS firms were assigned to the treatment groups in 60 percent of the prefectural cities ("95 percent prefectures"), and 70 percent of the CEMS firms were assigned to the treatment groups in the other 40 percent of the prefectural cities ("70 percent prefectures").¹³ This "double randomization" design allows us to causally identify the general equilibrium effects of pollution appeals by comparing the violation rates and emissions of firms across the 95 percent and 70 percent prefectures, conditional on their treatment status. We also test whether ambient air pollution concentrations were equal in the 95 percent and 70 percent prefectures, which provides information on whether there are decreasing returns to scaling up participation.

B. Experimental Implementation

The experimental period started on May 6, 2020 and ended on December 31, 2020. There are three key steps: (i) identify and verify CEMS firms that violated the emission standards based on the data in the previous 24 hours; (ii) file appeals through different channels according to the experimental design; and (iii) document government responses. Here, we provide some more detail on each of these steps.¹⁴

For the first step, we combined an algorithm that we developed with human judgment to mimic the definition of pollution violations set by the Ministry of Ecology and Environment.¹⁵ In real time, the algorithm identified all firms violating the national emission standards, based on their average pollution emission concentrations in the previous 24 hours.¹⁶ If a CEMS plant is being monitored for multiple pollutants, we identify it as a violator if any of the pollutants violate the emission standard. We then employed 12 environmental science graduate students

¹³These numbers are chosen so that each of the six treatment arms and the control arm will be similar in terms of sample size, which maximizes the statistical power in pairwise tests across different arms.

¹⁴A few implementation details prior to the study period are worth mentioning. In January 2020, we collected the phone number of all the firms in the sample and the official Weibo account of every local EPA. Between January and March 2020, we trained research assistants to identify and verify violations of emissions standards and trained citizen volunteers to file appeals via different channels following the experimental assignments. Additionally, we conducted a small-scale pilot in April to ensure all the research assistants could complete the daily tasks on time.

¹⁵The MEE was established in 2018 in replacement of the MEP.

¹⁶For each pollutant being monitored, the MEE determines a specific emission standard for each CEMS plant. Online Appendix Figure A2 plots the distribution of SO₂ and COD emissions standards across all CEMS plants in our sample.

to manually double-check the violations identified by the algorithm. Because the CEMS equipment often continues to run after production is suspended and there is little air/water flow, emissions concentrations can be abnormally high for a short period of time. To remove these false positives, the graduate students identified the violations that occurred after production had stopped or that were due to mechanical spikes by examining complementary indicators such as other pollutants concentrations and water/gas flows. Once the students, who were blinded to treatment assignment, identified true violations of the emissions standards, they took screenshots from the CEMS web pages (see online Appendix B for details).

The second step was to use the verified pollution violations and file appeals. Every day, we generated the list of verified violations, produced the script used for each appeal, and determined the delivery method based on the firms' treatment assignments. These appeals were eventually filed by a group of citizen volunteers, whom we recruited through three environmental NGOs. The volunteers who made appeals did not reveal information about their background or organizational affiliation, such that the appeals mirrored how the public normally participates.

We gave each volunteer no more than 15 appeals per day to provide enough time to follow the protocols (e.g., reporting by phone is limited to the working hours of the recipient). To avoid repetitive appeals, if a CEMS firm committed consecutive violations spanning multiple days, we waited until the next week before filing a second appeal. Online Appendix B describes the implementation protocol in greater detail and provides sample screenshots of appeals in each arm.

An additional part of the second step was the amplification of half of the public Weibo appeals. For the T2B arm, we hired a social media promotion company to boost the publicity of the appeals. Specifically, the company added roughly 10 "likes" and "shares" to the Weibo appeals (T2B), using a variety of existing and active company-operated Weibo accounts.

In the third step, the volunteers tracked the responses to the appeals from the local governments, which primarily came back as Weibo direct messages, Weibo public replies, 12369 phone calls, and 12369 website replies. We recorded the timing and content of each government response and matched them to the corresponding appeal.

Table 1 summarizes some basic facts about the experiment. First, as mentioned, we include the universe of 24,620 CEMS firms in China's 333 prefectures. During the eight-month treatment period, other citizens, who were not part of the research team, filed a total of 271,859 pollution appeals to the government; 5,478 were explicitly about pollution violations committed by the CEMS firms.¹⁷ Most appeals did not target specific CEMS firms and were instead about some unpleasant odors or dirty waters that people encountered near their communities. Second, the experiment was conducted in collaboration with the three NGOs, which collectively organized 15 volunteers on an average day to file pollution appeals. During the study period, a total of 120 Weibo accounts were used by the volunteers to file pollution appeals. We also hired 12 environmental science graduate students to

¹⁷These numbers are based on the administrative data covering the universe of citizen appeals filed in 2020, which we obtained from the MEE. We excluded pollution appeals filed by the research team when calculating these numbers.

TABLE 1—SUMMARY OF BASIC FACTS ABOUT THE EXPERIMENT

(1)	(2)	(3)	(4)
<i>Panel A. Experimental environment</i>			
Number of prefectures covered	Number of CEMS firms covered	Number of appeals filed by other citizens during the experiment	Number of appeals filed by other citizens about CEMS firms during the experiment
333	24,620	271,859	5,478
<i>Panel B. Experimental team</i>			
Number of partnering NGOs	Number of citizen volunteers on any given day	Number of Weibo accounts involved in appeals	Number of environmental science graduate students verifying violations
3	15	120	12
<i>Panel C. Experimental implementation</i>			
Number of violations during experiment according to CEMS raw data	Number of violations during experiment verified by research team	Number of appeals filed by research team	Number of formal responses to appeals filed by research team
12,596	5,366	2,941	1,161

Notes: This table reports the background and the implementation of our experiment. Our experiment started on May 6, 2020, and ended on December 31, 2020. Information on the number of appeals filed by other citizens is obtained from the MEE's administrative record, which we matched to the CEMS sample. The 3 partnering NGOs, which prefer to remain anonymous, helped us recruit and organize citizen environmental volunteers, maintaining 15 individuals ready to file appeals on any given day. The number of verified appeals is lower than the number of appeals in the CEMS raw data since we were conservative and excluded cases that might be driven by outliers or mechanical errors. The number of appeals filed is lower than the total number of violations verified by the research team because we did not appeal against violations committed by the control firms, nor did we file appeals repeatedly within a week about the same firm.

verify pollution violations. Third, the CEMS raw data showed 12,596 pollution violations during the study period, but our verification process revealed that only 5,366 violations were real, which were committed by 2,363 different CEMS firms.¹⁸ We filed 2,941 appeals according to the treatment assignments; for the remaining 2,425 violations, appeals were not filed either because these violations were committed by the control firms or because we did not file multiple appeals within a week about the same firm's violations. Our appeals generated 1,161 formal responses from the regulators.

C. Ethical Considerations

Prior to implementing this experiment, we carefully considered the ethical implications of working with a partner nongovernmental organization to file appeals. While online Appendix C discusses the ethical considerations in more detail, several points are worth highlighting. First, the rights of citizens to make appeals against violations is legally protected and explicitly encouraged by national policies. All local governments are mandated to operate the multiple channels of making appeals that we study in this experiment. Second, we consulted with several nongovernmental organizations that already had multiple years of experience filing appeals and were not advised of any repercussions to their staff or organizations. Third, we worked with a nongovernmental partner that was already active in

¹⁸ Here, if a firm simultaneously violated the emissions standards in multiple stacks or for multiple pollutants, we count that as one daily violation. Consecutive violations spanning multiple days are also combined as one violation. As a result, the violation rate in Table 1 is substantially smaller than the violation rate indicated by Figure 1.

environmental monitoring, so the treatments were not outside of the existing scope of their work. Fourth, we were in daily contact with our partner organization and never learned of any adverse events or pressures in response to the appeals. Finally, because we did not collect data from or about any individual people, two separate IRBs in the United States determined that this study was not considered research with human subjects.

D. Data, Balance Tests, and Empirical Description of the Treatments

The analysis is conducted on a dataset that results from combining several sources of information (Buntaine et al. 2024). These MEE data cover all the CEMS firms and include information on firm name, social credit code, industry, main pollutant type, hourly emission concentrations of various pollutants, hourly gas and water flows, pollution violation status, among other measures. In 2020, due to the COVID-19 lockdown, most CEMS firms suspended production until the economy reopened in mid-March, so we drop the first ten weeks of the 2020 CEMS data from the sample.¹⁹ The official database of CEMS firms was matched to our experimental data, including treatment arm, specific appeals, and government responses.²⁰

Two other government datasets were critical ingredients. First, we merged in MEE data on all citizen appeals against the CEMS firms in 2020 made either through the 12369 website or by phone. This database of appeals includes appeals filed by other citizens on their own volition and the ones generated by the experiment. Second, we also merged in the administrative data on firm registrations. These data contain information on the date of establishment, industry, business address, business type, registration status, and other measures. The firm registration data are merged with the CEMS data using the social credit code.

Table 2 reports on balance tests across the experimental arms. In column 1, we present the mean and standard deviation of the control group, for variables such as the share of firms in different industries; the total amount of pollution penalties paid in the previous year; frequent violators in the previous year; and various measures of pretreatment environmental performance (in the eight weeks before the treatment began), including the severity of violations and emission concentrations. We then compare each treatment arm to the control arm, implemented by running a regression of each outcome variable on a set of treatment dummies. In columns 2 to 6, we present the regression coefficients and standard errors for each variable-arm combination. As we can see, the treatment arms are well balanced with the control arm along almost all dimensions, confirming that our randomization was well executed. Online Appendix Table A1 also reports the detailed breakdown of industries by experimental arm.

¹⁹From mid-January to mid-March 2020, China was struck by COVID-19, and many CEMS firms suspended their production due to compulsory lockdowns. By late March 2020, almost all Chinese cities reopened because COVID-19 was already deemed under control. During the experimental period, production resumed, and firms operated as usual.

²⁰For analysis, we use the official CEMS data provided annually by MEE rather than the data published daily on the provincial government websites used to identify violations. The official data from MEE are more complete and have been cleaned of basic errors that occasionally appear in the real-time data.

TABLE 2—BALANCE TEST

	Private appeals					Public appeals Weibo T2-C
	Control C	Messaging T1A-C	Website T1B-C	Call Gov T1C-C	Call firm T1D-C	
	(1)	(2)	(3)	(4)	(5)	
<i>Panel A. Outcomes</i>						
SO ₂ violations	0.217 (2.202)	0.011 (0.052)	0.030 (0.072)	0.022 (0.052)	0.052 (0.058)	0.082 (0.078)
COD violations	0.095 (0.862)	0.014 (0.023)	0.006 (0.023)	0.001 (0.020)	0.017 (0.024)	0.036 (0.025)
Total violations	0.739 (4.927)	0.084 (0.125)	0.034 (0.128)	0.000 (0.128)	0.120 (0.136)	0.181 (0.156)
SO ₂ concentrations	135.2 (982.0)	−21.7 (21.6)	−14.5 (15.8)	−19.1 (22.9)	−37.0 (33.5)	−8.4 (18.2)
COD concentrations	57.6 (69.1)	1.3 (2.2)	3.8 (3.1)	1.8 (3.5)	2.0 (2.4)	0.6 (3.6)
Gas penalty	0.008 (0.146)	0.001 (0.004)	−0.003 (0.003)	−0.000 (0.002)	0.002 (0.004)	−0.003 (0.003)
Water penalty	0.001 (0.055)	0.001 (0.002)	0.002 (0.002)	0.003 (0.002)	0.002 (0.002)	0.002 (0.002)
Total penalty	0.009 (0.156)	0.002 (0.004)	−0.002 (0.003)	0.003 (0.003)	0.004 (0.004)	−0.001 (0.003)
Frequent violators	0.056 (0.229)	−0.008 (0.007)	0.004 (0.009)	0.009 (0.007)	0.005 (0.006)	0.006 (0.008)
<i>Panel B. Industries</i>						
Mining industry	0.024 (0.154)	0.001 (0.004)	−0.004 (0.006)	0.001 (0.005)	−0.002 (0.004)	0.003 (0.006)
Manufacturing and power plants	0.730 (0.444)	0.015 (0.012)	0.018 (0.015)	0.021 (0.014)	0.012 (0.011)	0.019 (0.015)
Sewage treatment	0.166 (0.372)	−0.017 (0.010)	−0.015 (0.012)	−0.007 (0.013)	−0.007 (0.010)	−0.012 (0.013)
Others	0.080 (0.272)	0.001 (0.007)	0.001 (0.010)	−0.015 (0.010)	−0.003 (0.007)	−0.010 (0.009)

Notes: This table reports balance tests across different experimental arms using data from the pretreatment period. For outcomes on pollution concentrations and violations, the sample includes eight weeks before the start of the experiment. For pollution penalties, the sample is from 2019. For frequent violators, we define a firm as a frequent violator if it violated more than ten times in 2019. Column 1 reports the means and standard deviations of the control arm. Columns 2–6 report the difference between each appeal arm and the control arm. We control for province fixed effects. Standard errors are clustered at the prefecture level.

III. Empirical Results

This section presents the baseline results from estimating the following econometric model:

$$(1) \quad Y_{ijt} = \sum_j \alpha_j T_{ij} \times Post_t + \gamma_i + \eta_t + \epsilon_{ijt},$$

where Y_{ijt} is the outcome of interest for firm i , assigned to arm j , on day t . T_{ij} represents the randomly assigned arm of firm i , and it is interacted with $Post_t$, which is a dummy variable that equals one after the experiment commenced.²¹ We control for firm fixed effects, γ_i , and day fixed effects, η_t . Since we cross-randomize treatment intensity at the prefecture level, we also estimate more saturated specifications

²¹ The week of May 7 is the eighteenth week of the year. Since the first ten weeks are excluded from our sample due to COVID-19 lockdown, the pretreatment period corresponds to the first seven weeks in our sample.

TABLE 3—POLLUTION APPEALS AND FIRM VIOLATIONS/EMISSION CONCENTRATIONS

	Violation (1a)	Violation (1b)	SO ₂ (2a)	SO ₂ (2b)	COD (3a)	COD (3b)
Private appeals ($T1 \times Post$)	−0.003 (0.001)	−0.002 (0.001)	−5.6 (3.6)	−5.9 (3.6)	−0.3 (0.9)	−0.4 (0.8)
Public appeals ($T2 \times Post$)	−0.006 (0.002)	−0.006 (0.001)	−15.8 (4.4)	−16.2 (4.4)	−2.1 (1.2)	−2.2 (1.2)
$H_0: T1 < T2$	$p = 0.01$	$p = 0.00$	$p = 0.01$	$p = 0.02$	$p = 0.03$	$p = 0.03$
Control mean	0.009	0.009	132.5	132.5	59.1	59.1
Control SD	0.096	0.096	539.5	539.5	78.8	78.8
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Day fixed effects	Yes		Yes		Yes	
Province-by-day fixed effects		Yes		Yes		Yes
Observations	7,100,881	7,100,881	2,216,208	2,216,208	2,459,622	2,459,622

Notes: This table reports the regression results from estimating equation (1). In columns 1a and 1b, we use firm-day-level data, and the outcome variable is a dummy variable that equals one if the firm violates an emission standard on that day, and zero otherwise; in columns 2a and 2b, we use pipe-day-level data, and the outcome variable is the daily average emission concentration of SO₂ (mg/m³); in columns 3a and 3b, we use pipe-day-level data, and the outcome variable is the daily average emission concentration of COD (mg/L). For each outcome, in the column “a,” we control for firm fixed effects and day fixed effects; in the column “b,” we control for firm fixed effects and province-by-day fixed effects. Standard errors are clustered two-way by prefecture and week.

that include province-by-day fixed effects to control for time-varying differences in regional enforcement. Standard errors are clustered two-way by prefecture and week.

The α_j 's are the parameters of interest. They measure the causal effects of different types of appeals relative to the controls during the experiment. In this section, we report on specifications that estimate the average effects of private appeals (T1A–T1D combined) and public appeals (T2A and T2B combined). We examine specific treatment arms when analyzing the underlying mechanisms. These estimates represent the “intention to treat (ITT)” effects since the assigned appeal type is only triggered after the firm commits a violation during the experimental period.²²

A. Pollution Appeals and Environmental Performance

Table 3 summarizes the results from the estimation of two versions of equation (1) for three measures of environmental performance. For each outcome, the column “a” specification includes firm and day fixed effects, and the column “b” specification replaces the day fixed effects with province-by-day fixed effects. In columns 1a and 1b, the outcome is $violation_{ijt}$, which is a dummy variable indicating whether firm i committed any pollution violation on day t . In the next two pairs of columns, the outcome variables are the firms' daily emission concentrations of SO₂ for air pollution, and COD for water pollution. These two pollutants have the highest coverage for CEMS firms and are the most high-stakes “criterion pollutants” for evaluating the

²²To quantify the treatment effect of appeals, we also adjust the baseline specification with an instrumental variable (IV) approach, where whether a firm has received an appeal of a certain type is instrumented by “*treatment assignment* $\times$ *whether experiment has started*.” The procedure and results are displayed in online Appendix D and online Appendix Table A14.

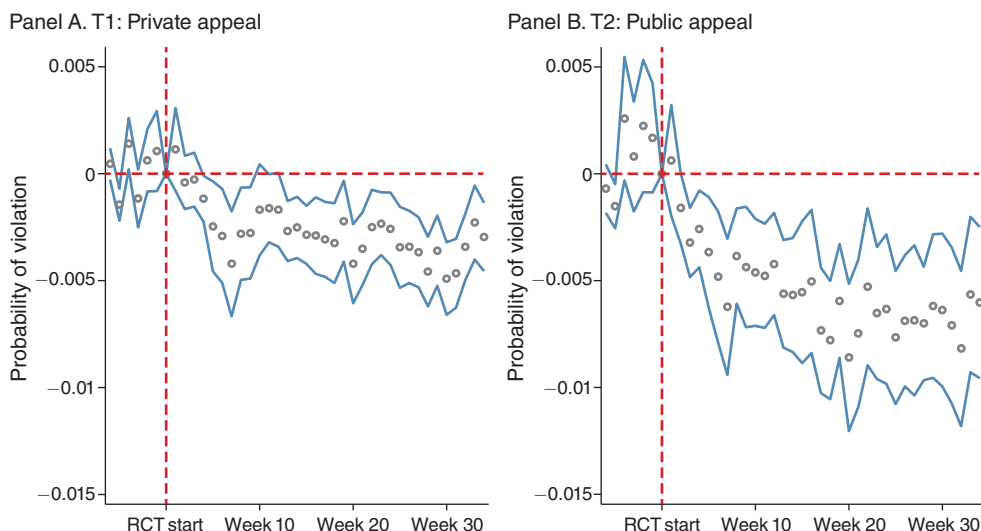


FIGURE 3. EVENT STUDIES

Note: This figure presents coefficients and 90 percent confidence intervals on Treatment $\times$ Week interactions from regressions of violation on Treatment Week, firm fixed effects, and week fixed effects. Standard errors are clustered two-way by prefecture and week.

environmental performance of local government officials (He, Wang, and Zhang 2020).

Violations.—The results indicate that appeals greatly reduce violations, especially public appeals. In the preferred column 1b specification, the private appeals treatment (T1) reduces the probability of a daily violation by 0.227 percent, relative to the control group; this is about 24 percent of the control group's mean of 0.936 percent. The public appeals treatment (T2) decreases the probability of a violation by a larger margin: roughly 62 percent relative to the control group's mean. The table also documents that the difference in magnitudes is statistically significant, as the null hypothesis that the public appeals treatment is smaller in magnitude than the private one is rejected at conventional significance levels. Finally, we note that the results are quantitatively similar across the two specifications.²³

²³ We conduct additional checks in the online Appendix. First, in Table 3, we defined pollution violations based on whether the monitored emission concentration exceeded the standard value set by the MEE. However, it is possible that some of these monitored values are driven by mechanical errors or production suspensions instead of actual pollution violations. In online Appendix Table A2, we refine the definition of pollution violations to exclude cases with minimal levels of measured airflows, as these may be instances when the plant is not operating. The results with this alternative definition of a violation are qualitatively the same as those in Table 3, confirming the powerful effect of the public appeal treatment at reducing violations and the more modest effect of private appeals. Second, in online Appendix Table A3, we report alternative standard errors by clustering at either the prefecture level or at the prefecture-by-arm level, and if anything, the statistical significance increases under these specifications. Third, in online Appendix Table A4, we aggregate the data to either the firm-month level or the firm-week level and run Poisson regressions with the same set of baseline fixed effects, and the main results still hold.

Figure 3 provides an opportunity to understand how the treatment effects evolve over time. Specifically, it reports the results from fitting a version of the column 1a specification where the treatment and *Post_t* interactions are replaced with treatment and week interactions. For both the private and public treatments, nonzero treatment effects emerge within a couple of weeks of the experiment's initiation. It is sensible that they do not appear immediately because the appeals are not initiated until a firm commits a violation. By week 20, it appears that both the T1 and T2 treatment effects have stopped increasing and stabilized at a level larger in magnitude than reported in Table 3 (which is an average over the entire experiment); this pattern suggests that the table's estimates understate appeals' long-run potential to reduce the incidence of violations. In addition, this finding also implies that firm and regulator adjustments are not simply a temporary reaction to a new stimulus. In online Appendix Figure A3, we also disaggregate T1 and present the trends of each sub-treatment arm, which show similar patterns. Moreover, it is reassuring that there is no evidence of a treatment effect in the period before the RCT began for either the private or public appeal arms, confirming that the randomization was well executed. Overall, these figures suggest that the long-run partial equilibrium effects of the treatments are larger than is reported in Table 3.²⁴

Emissions.—Columns 2 and 3 of Table 3 show that the public appeals treatment (T2) caused substantial reductions in air and water pollution emissions concentrations, while the private treatments (T1) led to much more modest reductions. The results from the preferred 2b and 3b specifications reveal that public appeals reduced the average firm's daily average SO₂ emission concentration by 16.2 $\mu\text{g}/\text{m}^3$, and its daily average COD emission concentration decreased by 2.2 $\mu\text{g}/\text{L}$; these are 12.2 percent and 3.7 percent declines from the control group's SO₂ and COD emission concentration levels, respectively. Further, the estimates also indicate that the private appeals treatments slightly reduced these two measures of air and water emissions concentrations, but the effects are much smaller and would not be judged to be statistically significant by conventional criteria.²⁵

Figure 4 further examines where in the distribution of emissions concentrations the Table 3, columns 2 and 3 estimates of public appeals come from. We report the results from fitting the preferred version of equation (1) with province-by-day fixed effects separately on indicators for whether a firm's daily SO₂/COD emissions concentration was 0–40 percent of the corresponding emissions standard, 40–80 percent, 80–100 percent, 100–200 percent, and greater than 200 percent. As we can see, under public appeals, firms' emissions concentrations became less likely to exceed the emission standards, which is especially true for extreme violations where their emissions concentrations more than doubled the national standards. Interestingly, these shifts in emissions concentrations appear to be inframarginal: the T2 firms did

²⁴That said, we acknowledge that the time frame of the experiment remains relatively short, and the regulators and firms might adjust differently in the much longer run.

²⁵Online Appendix Figure A4 is constructed identically to Figure 3 together with online Appendix Figure A3 and reports on how the emissions concentration treatment effects evolve over time. Here, too, the treatment effect grows over time, especially for the private appeals, presumably as violations cause the polluters to learn about the experimentally induced increase in scrutiny. The takeaway is that, holding constant the regulatory environment, the treatments' long-run equilibrium effects on emissions are larger than those reported in Table 3.

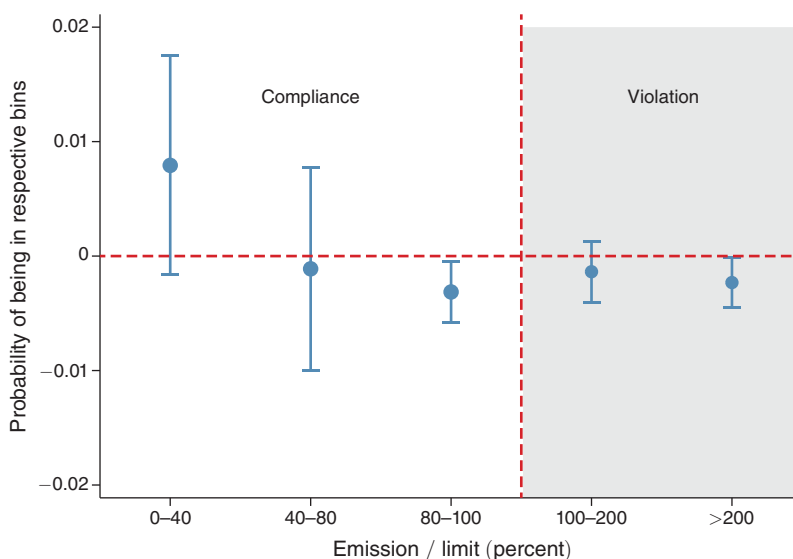


FIGURE 4. EFFECTS OF PUBLIC APPEALS ON EXCESSIVE VIOLATIONS

Notes: In this figure, we visualize how public pollution appeals shift the distribution of emission concentrations. We divide each firm's SO_2 (COD) emission concentration on a given day by the SO_2 (COD) emission limit for this firm set by the MEE and generate five bins based on this standardized emission variable. We regress the dummy variable for each bin on our treatment variables, using the same baseline specification in equation (1), and plot the coefficients and 90 percent CIs from these regressions. We control for firm fixed effects and province-by-day fixed effects. Standard errors are clustered two-way by prefecture and week.

not fall in the “barely compliant” bin (80–100 percent) more frequently. Instead, they became much more likely to fall in the “highly compliant” bin, where their emissions concentrations were below 40 percent of the national standards. This is consistent with the qualitative observation that most pollution violations committed by CEMS firms were driven by discontinued operations of their abatement facilities (to reduce energy use), and once these facilities become properly functioning, most CEMS firms' emissions concentrations can fall well below the emissions standards.

Additionally, we investigated whether the reduction in average emissions concentrations translated into reductions in total emissions, which are the relevant metrics for determining the impact on individuals' exposure to ambient pollution.²⁶ The impacts on emission concentrations and total emissions could differ if the treatments cause the plants to change their intensity of operations (e.g., a decline in concentration could be offset by an increase in the number of hours of operation or running the plant at full capacity more frequently during operating hours). While the CEMS data do not directly report hourly data on total emissions, we infer the changes in total emissions by investigating the average hourly gas/water flows recorded by the CEMS; the product of these variables and the emission concentrations equal average total emissions per hour for each pollutant. Online Appendix Table A5 reveals that

²⁶It is common for regulators around the world to focus on emissions concentrations, even though it is total emissions that matter for human health. For example, regulators in India and the United States focus on enforcing emissions concentration standards.

none of the treatments had a meaningful impact, either economically or statistically, on the flow of pollutants. We thus conclude that there were reductions both in emissions concentrations and total emissions.

B. Heterogeneity in Treatment Effects, Based on Firm Characteristics

Table 4 tests for heterogeneity in the treatment effects based on whether firms are state-owned enterprises (SOEs), produce “final” (rather than “intermediate”) goods that might make them more concerned about their public image,²⁷ and whether they committed violations during the seven-week pretreatment period. The state-owned enterprise tests are reported in panel A, columns 1–3 for the violation rate, SO₂ emissions concentration, and COD emissions concentration, respectively. The analogous regression results for the other two categories are reported in panel B and panel C. Throughout the table, we report the results from the more demanding specification that includes firm and province-by-day fixed effects.

A salient finding in Table 4 is that the overall treatment effects are driven by the firms that committed violations during the seven weeks prior to the experiment (panel C). Indeed, there is little evidence of any treatment effect among the subsample of firms that did not violate in the pretreatment period.²⁸ Additionally, the treatment effects for the SOEs are larger in magnitude than for private firms, although the difference is not statistically significant by conventional criteria. Finally, there is little consistent evidence that the treatment effect is different for final goods firms, compared to the rest of the sample, indicating that “corporate social responsibility” concerns are unlikely to be the driving force behind the main results. More broadly, this finding fails to support the possibility that the public appeals treatment effects are due to firms’ preemptive response to these appeals (as opposed to responses driven by regulators’ actions).

IV. What Explains the Estimated Private and Public Appeals Treatment Effects?

This section explores several explanations for the estimated private and public treatment effects observed in the previous section. Section IVA reports on our efforts to understand the source of the private appeals treatment effect and ultimately concludes that it is difficult to infer much about the mechanisms that explain the private appeals’ modest effects. Section IVB presents a variety of evidence that together suggests that the estimated public appeals treatment effects are large, because they tilt the balance of local regulators’ often competing goals away from facilitating economic growth and toward avoiding public unrest due to pollution. Section IVB then details a series of other potential explanations that are contradicted by the data.

²⁷ This variable is defined based on industry code in the business registration data.

²⁸ The amount of frequent and nonfrequent violators is balanced across different experimental arms. This heterogeneity could be mechanical, given that the frequent violators have more room for improvement. Another possibility is that frequent violators might be more responsive to citizen appeals. Comparing how frequent and nonfrequent violators’ emissions changed after they received their first appeals in our experiment, we find no evidence of differential responsiveness, thus suggesting that the heterogeneity might be mechanically driven by the difference in baseline violation rates.

TABLE 4—HETEROGENEITY ANALYSES ON FIRM VIOLATIONS AND EMISSION CONCENTRATIONS

	Violation (1)	SO ₂ (2)	COD (3)
<i>Panel A. Ownership</i>			
Private appeals ($T1 \times Post$)	−0.002 (0.001)	−6.9 (4.0)	−0.3 (1.0)
Public appeals ($T2 \times Post$)	−0.005 (0.002)	−19.4 (5.1)	−2.3 (1.4)
Private appeals ($T1 \times Post$) $\times$ SOE	−0.003 (0.004)	8.2 (15.0)	0.6 (2.7)
Public appeals ($T2 \times Post$) $\times$ SOE	−0.005 (0.005)	24.4 (17.0)	1.9 (2.9)
$Post \times SOE$	0.003 (0.004)	−6.9 (6.3)	0.1 (2.3)
Observations	6,016,662	1,971,513	2,128,655
<i>Panel B. Product</i>			
Private appeals ($T1 \times Post$)	−0.002 (0.001)	−7.1 (4.4)	−0.1 (1.0)
Public appeals ($T2 \times Post$)	−0.006 (0.002)	−18.9 (5.3)	−2.2 (1.6)
Private appeals ($T1 \times Post$) $\times$ Final	−0.002 (0.002)	7.8 (4.8)	−1.1 (1.5)
Public appeals ($T2 \times Post$) $\times$ Final	0.002 (0.003)	10.1 (5.9)	−0.3 (2.5)
$Post \times Final$	0.002 (0.002)	−3.2 (4.9)	2.7 (1.3)
Observations	6,016,662	1,971,513	2,128,655
<i>Panel C. Pretreatment Violations</i>			
Private appeals ($T1 \times Post$)	−0.000 (0.001)	−7.9 (5.1)	0.2 (0.8)
Public appeals ($T2 \times Post$)	−0.001 (0.001)	−15.7 (5.5)	−2.0 (1.6)
Private appeals ($T1 \times Post$) $\times$ Frequent	−0.013 (0.006)	9.4 (7.7)	3.0 (2.7)
Public appeals ($T2 \times Post$) $\times$ Frequent	−0.030 (0.009)	−2.0 (12.8)	4.9 (3.2)
$Post \times Frequent$	−0.028 (0.007)	−5.7 (4.6)	−5.1 (2.7)
Observations	5,827,579	1,887,624	1,998,089
Firm fixed effects	Yes	Yes	Yes
Province-by-day fixed effects	Yes	Yes	Yes

Notes: This table reports the results for heterogeneity analyses. In column 1, we use firm-day-level data, and the outcome variable is a dummy variable that equals one if the firm violates an emission standard on that day, and zero otherwise; in column 2, we use pipe-day-level data, and the outcome variable is the daily average emission concentration of SO₂ (mg/m³); in column 3, we use pipe-day-level data, and the outcome variable is the daily average emission concentration of COD (mg/L). “SOE” is a dummy variable that equals one if the firm’s majority shareholder is the government. “Final” is a dummy variable indicating whether the firm produces final good instead of intermediate good based on its industry code. “Frequent” is a dummy variable that equals one if the firm committed pollution violations in the seven weeks prior to the experiment. We control for firm fixed effects and province-by-day fixed effects. Standard errors are clustered two-way by prefecture and week.

A. Private Appeals

Online Appendix Table A6 explores the mechanisms that underlie the overall T1 treatment effect by disaggregating the effects into several treatment arms. There

are three observations. First, there is little evidence that private appeals to the government are more effective than private appeals to firms, as we cannot reject the null that the effects of T1A, T1B, T1C, and T1D are equal. Second, there is also no evidence that the channel through which the private appeals are delivered matters, as we cannot reject the null that T1A, T1B, and T1C are equal. Finally, privately informing both the government and the firm has little additional effect on the outcomes because the coefficient associated with the interaction of the T1C and T1D treatments is statistically insignificant. In other words, there are no complementarities in privately appealing to the regulator and to firms, implying that publicity is not operating exclusively through firm-side changes.

Overall, the magnitude of the private appeals treatment effect is unaffected by whether the firm or regulator receives the appeal and by the medium through which the appeal is delivered. It is therefore difficult to infer much about the mechanisms that explain the private appeals' modest effects.

B. Public Appeals

Regulators' Political Incentives.—Section IC provided qualitative evidence that regulators face the competing incentives of facilitating economic growth and reducing environment-related social unrest. These incentives mean that in the absence of credible threats for collective actions, it is natural to presume that regulators would frequently choose to turn a blind eye to pollution violations committed by important local firms to foster higher economic growth. However, failures to prevent collective actions could veto the regulators' promotion cases; this possibility gives the regulators high-powered incentives to respond swiftly to public attention about a pollution violation with significant regulatory effort.

We designed one of the treatments to assess the importance of "potentially explosive public sentiment" in driving regulatory actions. Recall, we created additional experimental variation in the visibility of public pollution appeals by randomly assigning additional "likes" and "shares" for half of the T2 public appeals posted on Weibo. In total, T2B received approximately 10.56 "likes" and "shares," compared to 0.66 for T2A. As we noted above, this is more than three times greater than the average number of likes/shares from posts by the environmental NGO and the local environmental regulator and at the ninety-ninth and ninetieth percentiles of the distributions of those organizations' tweets, respectively. Thus, we believe this treatment will elevate regulators' assessment that the underlying incident could cause social unrest.

To empirically test the impact of this treatment, we compare the regulatory effort in response to public appeals between these promoted public appeals (T2B) and public appeals that were not promoted (T2A). For this analysis, the sample is restricted to firms assigned to public appeals (T2). Table 5 reports the results of fitting an equation where the outcome is a measure of regulator response to a public appeal on Weibo. The explanatory variable of interest is an indicator for the T2B treatment that involves the appeal's promotion. This is a cross-sectional regression, and the unit of observation is a pollution appeal that we posted on Weibo regarding a firm's violation. Thus, the results capture the variation due to the combined effect

TABLE 5—SOCIAL MEDIA PUBLICITY AND GOVERNMENT RESPONSIVENESS

	Whether respond	Whether respond	Response length	Response length	On-site audit	On- site audit
	(1a)	(1b)	(2a)	(2b)	(3a)	(3b)
Visibility promotion (<i>T2B</i>)	0.06 (0.03)	0.06 (0.03)	34.6 (13.4)	33.8 (13.4)	0.04 (0.02)	0.05 (0.02)
Control Mean	0.16	0.16	33.1	33.1	0.07	0.07
Control SD	0.36	0.36	117.9	117.9	0.26	0.26
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Province fixed effects		Yes		Yes		Yes
Observations	662	658	662	658	662	658

Notes: This table reports the regression results for public Weibo appeals on local government responsiveness. We use the sample of firms in the public Weibo appeal to government arm. The unit of analysis is each Weibo appeal. “Whether respond” is a dummy variable that equals one if the government replies to our Weibo appeal, and zero otherwise; “Response length” is the word count of the government’s Weibo reply to our appeal, which is counted as zero if there is no response; “On-site audit” is a dummy variable that equals one if the government replies to our Weibo appeal with proof of an on-site investigation, and zero otherwise. For each outcome, in the column “a,” we control for month fixed effects; in the column “b,” we control for month fixed effects and province fixed effects.

of social media posting and public promotion (*T2B*), relative to the effect of social media posting and no experimental public promotion (*T2A*). The dependent variables include an indicator for whether the Weibo appeal receives a response from the regulator (columns 1a and 1b); the response’s length measured in words, where nonresponse is coded as zero (columns 2a and 2b); and an indicator for whether the response includes proof of an on-site inspection or audit of the violator (columns 3a and 3b). The “a” specifications include fixed effects for the day of the violation, and the “b” specifications add province fixed effects.

It is apparent that environmental regulators are responsive to the publicity of appeals. The columns 1a–b results indicate that adding Weibo likes/shares significantly increases the probability that the regulator replies to the Weibo appeal by approximately 6 percentage points, which is a roughly 40 percent increase in the baseline response rate of 15.5 percentage points. The treatment also doubles the average length of the response (columns 2a–b) and increases the probability of a documented on-site inspection or audit by more than 60 percent. Overall, these results suggest that more publicity significantly increases local governments’ responsiveness and effort to regulate pollution, as well as indicating the power of social media networks to provide a forum to launch the publicity.

Since *T2B* was randomized at the *appeal* (instead of *firm*) level, the subset of firms that committed multiple violations ultimately received both *T2A* and *T2B*; it is not possible to measure impacts on firm-level outcomes like the paper does when measuring the impacts of *T1* and *T2*. That said, we still try to shed light on *T2B*’s impact on firm outcomes by comparing the subsequent violation patterns for firms that randomly received promoted versus non-promoted Weibo appeals for their *first* violations during our sample period. Online Appendix Table A8 demonstrates that firms whose first violation received a Weibo appeal with promotion, compared to firms whose first violation received a generic Weibo appeal, commit roughly 0.34 fewer violations during the experiment and are 8–9 percentage points less likely to commit any subsequent violation. These effects are about 50 percent and 25 percent of the mean among the public appeal only group, respectively, and

so are qualitatively large although they are marginally statistically significant. The magnitude is especially noteworthy because some of the “control” group received the “social media public promotion” for subsequent violations.²⁹

We further probe the hypothesis that political incentives explain the public appeal treatment effect. Following Guo (2009), we collected the résumés of all the prefectural environmental bureau chiefs in 2020 and calculated how many years they have left in their current five-year terms. Since most bureaucratic promotions happen after the completion of a term, regulators at the beginning of their terms (i.e., the first two years) tend to have weaker political incentives compared to their peers who are more advanced in their current terms (i.e., last three years). Panel A of online Appendix Table A7 finds that in years 3–5 of a regulator’s 5-year term, the likelihood of a firm violating 3 months prior to the experiment increased by 2.7 percent (see column 1b), which is a 67.5 percent increase, relative to the regulator’s first 2 years in office. In a companion result, panel B of online Appendix Table A7 reveals that both private and public appeals have a larger and statistically significant impact on firm daily violation rates in prefectures where the head environmental regulator is in years 3–5 of their term, compared to prefectures where they are in the first two years. These results support the hypothesis that local regulators’ career incentives influence the intensity of environmental enforcement and help to explain the public appeals treatment effects.

Alternative Explanations.—We investigated several other potential mechanisms for the public appeals results, but the data failed to support them. For example, there are often concerns about data quality in China, and one possibility is that polluting firms might respond to public appeals by manipulating the CEMS data rather than abating pollution. As explained in Section IB, the CEMS utilizes a series of technologies and follows strict protocols to ensure the accuracy of the data, which, in principle, leaves little room for firms to influence the automatic emission readings. Nevertheless, we investigated this possibility by comparing the frequency of suspicious readings across the experimental arms. The results in online Appendix Table A12 indicate that the experimental interventions had no impact on the probability that the CEMS technologies were operated for fewer than 20 hours in a day (ninetieth percentile) or the probability that the CEMS recorded unusually low emission concentrations (i.e., below 10 percent of its yearly average on a fully operating day), suggesting that data manipulation is unlikely to drive our main findings. Moreover, as we will discuss in detail in Section V, the findings on firm-level emission reductions are corroborated by changes in ambient pollution levels at the prefectural city level, further supporting the baseline findings that appeals, especially public ones, cause improvements in firms’ environmental performance.

Another possible explanation for the public appeals treatment effect is that they reduce the information asymmetry between local regulators and central government officials. A reduction in this asymmetry might incentivize local regulators to vigorously regulate violations to avoid sanctions or oversight from above (Anderson et al. 2019; Buntaine, Zhang, and Hunnicutt 2021b). This mechanism is unlikely in our context since any citizen appeal in the 12369 platform is automatically documented in

²⁹ Over the course of the experiment, the firms whose first violation received a promoted Weibo appeal received an additional 0.38 promoted Weibo appeals.

the MEE's central system, meaning the central government is immediately informed of violations, which leaves little scope for information asymmetries across different levels of government. However, we tested this possibility directly: in half of the private Weibo appeals (T1a), we randomly threatened the local regulator that "if the issue does not get resolved, we will bring it to the central government." The results in online Appendix Table A13 indicate that this treatment did not have a statistically meaningful effect on any of the three measures of regulatory effort, suggesting that concerns about central government oversight do not drive the baseline findings.³⁰

Finally, it is possible that the public appeals posted by the research team on social media inspired other citizens to file more *private* appeals about the same violation, which in turn triggered stronger regulatory responses. If this were the case, the stronger regulatory responses could be mainly driven by the accumulation of *private* appeals rather than the likelihood of a public appeal "blowing up" on social media. To examine this potential channel, we obtained the universe of 12369 private appeals data from the MEE and matched this information to each CEMS firm in the sample. We find that our interventions did not affect the number of private appeals filed by other citizens, and the null results are precisely estimated (online Appendix Table A9). Moreover, we investigate whether the promoted public appeals led to more private appeals from other citizens and find no such evidence (online Appendix Table A10). Relatedly, we also test whether increasing the number of private appeals about a pollution violation would affect violators' behaviors in the following months. Conditional on the severity of a violation, we cannot reject the null hypothesis that receiving multiple private appeals has the same effect as a single private appeal (online Appendix Table A11). Overall, these observational findings suggest that the effect of publicly appealing violations is not driven by crowding in other citizens' private appeals.

V. General Equilibrium Impacts of Pollution Appeals

Except for rare exceptions (e.g., Crépon et al. 2013; Egger et al. 2019), randomized control trials produce causal *partial equilibrium* estimates of an intervention but cannot provide evidence on the intervention's *general equilibrium* or indirect consequences. This setting is one where knowledge of the general equilibrium consequences may be especially important. This is because it is at least plausible that regulators responded to public and private appeals by shifting inspections and other regulatory effort between firms, allowing untreated firms to increase their emissions due to the reduced regulatory scrutiny.³¹ If appeals only shift enforcement, it

³⁰ Relatedly, we tested the possibility that the local officials were afraid of potential mainstream media exposure by randomly threatening to contact local newspapers about the violation. As shown in online Appendix Table A13, this also had no significant impact on regulatory effort, although some of these coefficients are imprecisely estimated. Since these randomly added threats of escalations (to media or to upper-level governments) signal higher levels of civil unrest, the lack of heterogeneity in online Appendix Table A13 also indicates that governments interpreting social media appeals as reflecting more anger is unlikely to be the main reason that public appeals are significantly more effective than private appeals.

³¹ This could be driven by either the regulators' competing goals of balancing economic growth and social stability (so that they allow more pollution for firms not at the center of public attention) or regulatory capacity constraints that limit the scope of regulation (so that regulating treated firms will reduce regulatory resources spent on control firms).

TABLE 6—GENERAL EQUILIBRIUM EFFECTS OF POLLUTION APPEALS

	Violation (1a)	Violation (1b)	SO ₂ (2a)	SO ₂ (2b)	COD (3a)	COD (3b)
<i>Panel A. Control group</i>						
<i>High Intensity</i> × <i>Post</i>	−0.003 (0.003)	−0.003 (0.003)	−1.5 (5.0)	−2.9 (3.5)	3.3 (2.0)	3.6 (1.9)
Observations	1,026,414	1,026,414	296,604	296,604	356,265	356,265
<i>Panel B. Treatment group</i>						
<i>High Intensity</i> × <i>Post</i>	−0.004 (0.002)	−0.003 (0.001)	−1.7 (4.6)	−2.3 (5.2)	1.0 (0.9)	0.9 (0.9)
Observations	6,074,467	6,074,467	1,919,513	1,919,513	2,103,337	2,103,337
Control mean	0.009	0.009	132.6	132.6	59.1	59.1
Control SD	0.096	0.096	539.5	539.5	78.8	78.8
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Day fixed effects	Yes		Yes		Yes	
Province-by-day fixed effects		Yes		Yes		Yes

Notes: This table reports the results of general equilibrium analyses. Panels A and B report the impact of assignment to the 95 percent prefecture group, relative to the 70 percent group, for the control and treatment groups. In columns 1a and 1b, we use firm-day-level data, and the outcome variable is a dummy variable that equals one if the firm violates an emission standard on that day, and zero otherwise; in columns 2a and 2b, we use pipe-day-level data, and the outcome variable is the daily average emission concentration of SO₂ (mg/m³); in columns 3a and 3b, we use pipe-day-level data, and the outcome variable is the daily average emission concentration of COD (mg/L). For each outcome, in the column “a,” we control for firm fixed effects and day fixed effects; in the column “b,” we control for firm fixed effects and province-by-day fixed effects. Standard errors are clustered two-way by prefecture and week.

is possible that they had little or even zero impact on total emissions and ambient pollution concentrations.

We designed the experiment to learn about the general equilibrium consequences of appeals by cross-randomizing treatment intensity across regions. Specifically, in 60 percent of the prefectural cities, 95 percent of the CEMS firms were assigned to the treatment groups, while in the other 40 percent of the prefectural cities, 70 percent of the CEMS firms were assigned to the treatment groups. We implement several tests, based on this cross-randomization, to assess the general equilibrium impacts.

Table 7 examines the impacts on ambient SO₂ pollution concentrations using data from national air quality monitoring stations in China, which are independent from the CEMS network and cannot be influenced by the CEMS firms. The entries come from the estimation of

$$(2) \quad SO2_{st} = \alpha \cdot High_s \times Post_t + \gamma_s + \eta_t + \epsilon_{st},$$

where $SO2_{st}$ is the average ambient SO₂ concentration recorded in prefecture s on day t ; $High_s$ is a dummy variable indicating whether prefecture s was experimentally assigned to the high-treatment-intensity group where 95 percent of the CEMS firms are assigned to one of the treatment arms; $Post_t$ is a dummy variable indicating whether day t is after the treatments were initiated; and γ_s and η_t are prefecture and day fixed effects, respectively. The coefficient of interest is α , which measures the effect of the 95 percent treatment, relative to the 70 percent treatment (there is not a control group of cities with 0 appeals). Standard errors are clustered two-way by prefecture and week. Finally, we note that we only examined SO₂ concentrations because

TABLE 7—POLLUTION APPEALS AND AMBIENT AIR POLLUTION LEVELS

	SO ₂ (1a)	SO ₂ (1b)
<i>High Intensity Region × Post</i>	−0.36 (0.20)	−0.37 (0.19)
Control mean	10.06	10.06
Control SD	6.59	6.59
City fixed effects	Yes	Yes
Day fixed effects	Yes	
Province-by-day fixed effects		Yes
Observations	90,603	89,443

Notes: This table reports the regression results using ambient SO₂ air quality data from more than 1,600 air quality monitoring stations in China. The unit of analysis is prefecture-day. In column 1a, we control for city fixed effects and day fixed effects; in column 1b, we control for city fixed effects and province-by-day fixed effects. Standard errors are clustered two-way by prefecture and week.

industrial production is responsible for more than 80 percent of China's total SO₂ emissions, while less than 50 percent of China's total COD emissions are from industrial sources, meaning that this outcome is unlikely to have sufficient statistical power.

The estimates displayed in Table 7 reveal relative reductions in ambient SO₂ concentrations in the high-intensity prefectures. Specifically, SO₂ concentrations decreased by more than 3.5 percent in the 95 percent prefectures, relative to the 70 percent ones. This finding, despite being noisy, is quite striking because ambient air quality measures have limitations for detecting the effect of the public and private appeal interventions.³² In addition to testifying to the far-reaching impacts of the nationwide interventions, the findings on ambient pollution also confirm that at a minimum, the baseline improvements in firms' environmental performance cannot be entirely explained by their manipulation of the CEMS data. Finally, we note that these results would lead to the rejection of the null hypothesis that there are countervailing general equilibrium forces that perfectly undo the treatment's partial equilibrium improvements in environmental performance documented in the preceding tables. In Figure 5, we see that the treatment effect on ambient SO₂ concentrations appears to increase over time, which is consistent with a positive general equilibrium effect as firms learn about increased public scrutiny, although statistical imprecision prevents definitive conclusions.

Table 6 returns to the firm data and estimates the impact of assignment to the 95 percent prefecture group, relative to the 70 percent group, on violations, SO₂ emissions, and COD emissions, with the aim of better understanding the treatments' general equilibrium impacts. Specifically, we estimate a version of equation (2) with daily observations on these firm-level outcomes. These regressions provide an opportunity to separately test for a general equilibrium response among control and treatment firms by assessing whether either of the following two opposing forces dominates: (i) limited regulatory resources or multitasking incentives that cause regulators to shift enforcement to the CEMS firms subject to appeals, and the firms not subject to the

³²For instance, ambient air quality measures are affected by other local emission sources (e.g., household coal consumption and non-CEMS polluting firms) and emissions from other jurisdictions because SO₂ can travel hundreds of miles. Additionally, changes in meteorological conditions can significantly influence ambient air quality.

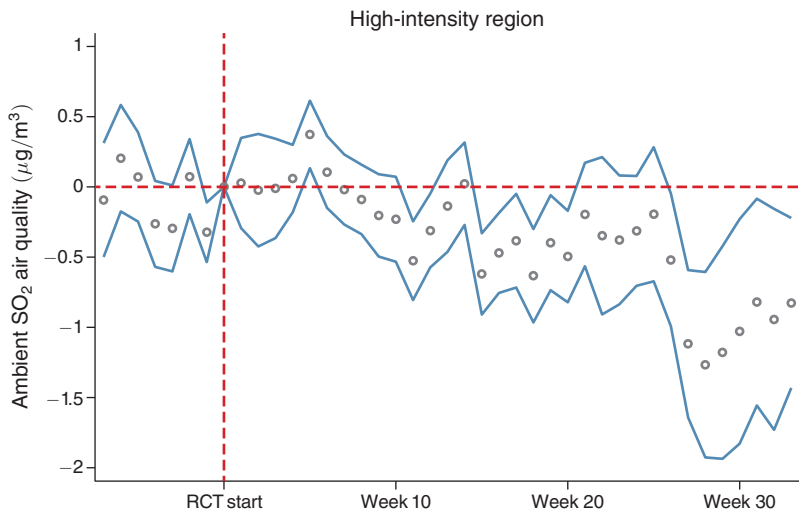


FIGURE 5. EVENT STUDIES FOR HIGH-INTENSITY REGION ON AMBIENT AIR POLLUTION LEVELS

Notes: This figure presents coefficients and 90 percent confidence intervals on High-Intensity $\times$ Week interactions from regressions of ambient SO₂ air quality on High-Intensity $\times$ Week, city fixed effects, and province-by-week fixed effects. Standard errors are clustered two-way by prefecture and week.

experimentally induced appeals responding by *increasing* their emissions, and (ii) some combination of a secular increase in enforcement or firms' response to a perceived increase in regulatory intensity causing these firms to *reduce* emissions.

In panel A, the sample is limited to control firms that were not directly affected by the treatment, meaning that this group provides a straightforward test of the net effect of these potential general equilibrium forces. The point estimates in column 1 suggest that control group violations did not vary significantly across the 95 percent prefectures and the 70 percent ones. The SO₂ and COD emissions estimates point in opposite directions, with neither being near statistical significance.

Panel B conducts the same exercise for the treatment firms. Here, too, in the absence of general equilibrium impacts, there is no reason for these measures of environmental performance to vary between the two groups of prefectures. The probability of a violation is 20 percent lower in 95 percent prefectures, relative to the 70 percent prefectures for the treatment firms, in the more robust column 1b specification; these estimates are statistically significant at the 10 percent level. There is also no evidence of a difference in average SO₂ or COD concentrations.

Overall, we conclude that the partial equilibrium treatment effects were not zero sum because the reductions among targeted plants were not undone by increased emissions from other plants. Indeed, it appears that the general equilibrium impacts might even be positive for the treatment firms, perhaps indicating that direct regulation and general deterrence are complementary.

VI. Conclusion

There are three main findings from this paper's nationwide field experiment in China that randomly appealed privately and publicly against pollution violations

through officially sanctioned channels. First, public appeals to the regulator through social media reduced violations by more than 60 percent and decreased air and water pollution (SO_2 and COD) concentrations by 12.2 percent and 3.7 percent, respectively. In contrast, private appeals caused more modest environmental improvements. Interestingly, the emissions reductions were concentrated among the plants that grossly exceeded the standard prior to the experiment rather than those just above the standard, and the violations reductions were concentrated among plants that frequently exceeded the standard prior to the experiment.

Second, the available evidence suggests that the public appeals are so effective because they tilt local regulators' often competing goals away from facilitating economic growth and toward avoiding pollution-induced public unrest. This conclusion is based on a qualitative examination of local regulators' career incentives and a series of quantitative tests. An especially important empirical finding is that experimentally increasing the visibility of social media appeals about a violation by adding likes/shares to the Weibo post greatly increased regulatory effort. In contrast, we fail to find evidence for several other potential mechanisms, including that the treatments caused firms to manipulate CEMS data.

Third, we find that the general equilibrium effects do not offset the partial equilibrium effects. If anything, they may even strengthen the partial equilibrium effects. This rare opportunity to assess the general equilibrium consequences of an experiment is based on randomly varying the share of firms subject to the treatments across China's 333 prefectures.

A complete cost-benefit analysis of pollution appeals is beyond the scope of this paper, but we can make a few observations about some key components of such a calculation. If publicly appealing every pollution violation generated the same response by firms as in this experiment, then there would be 51,000 fewer violations each year, a reduction of approximately 60 percent. This response would reduce China's total industrial SO_2 emissions by 9.2 percent and total industrial COD emissions by 2.9 percent relative to the baseline. A complete accounting of the resulting benefits would require reliable local air quality models to convert these emissions reductions into reductions in ambient air and water pollution and information on the willingness to pay for these improvements, but current pollution levies (1.26 RMB/kg for SO_2 , 1.4 RMB/kg for COD) provide a lower bound. At these rates, which are generally perceived as being too low, the projected emissions reductions are worth at least 360 million RMB per year.³³

On the cost side, we were unable to obtain data on the costs that firms incurred to reduce their emissions. However, it is worth noting that the marginal costs of citizens filing public appeals are very low—the average time used by our citizen volunteers to file an appeal in our experiment was less than five minutes. Based on the average hourly salary in China (20.1 RMB), the total labor cost of publicly

³³ China emits 3,954,000 tons of industrial SO_2 every year, roughly 75 percent of which is from the CEMS firms. So by reducing the CEMS firms' industrial SO_2 emissions by 9.2 percent, China's total industrial SO_2 emissions will fall by 272,826 tons. Similarly, China's yearly industrial COD emission is about 772,000 tons, roughly 75 percent of which is from the CEMS firms. So by reducing the CEMS firms' industrial COD emissions by 2.9 percent, China's total industrial COD emissions will fall by 16,791 tons. The total reduced pollution levies will therefore be $272,826 \times 1,000 \times 1.26 + 16,791 \times 1,000 \times 1.4 = 367,268,160$. This calculation is conservative because it ignores the potentially positive spillover effects as well as the positive impacts on other types of pollutants.

appealing all pollution violations in China would be around 170,000 RMB. Even if the costs of identifying violations and preparing appeals increased these costs substantially, the costs to the public of appealing are likely still orders of magnitude lower than the benefits to them of reduced pollution.

The paper has at least a few broader implications. First, it provides experimental evidence on the impacts of citizen complaints and appeals in environmental governance and underscores the power of social media in facilitating citizen involvement in enforcing policies in China. The results imply that social media provides strong signals of public demand for stringent enforcement, which in turn prompts regulators to recalibrate their approach to the trade-offs involved with environmental regulation.

Second, it deepens our understanding of how governments, firms, and citizens interact in China's local governance system. It shows that regulators use participation and the information it reveals about public discontent to gauge the value of imposing costly regulations on firms, and particularly so when lax enforcement has the potential to generate publicity. In addition, it demonstrates that the failure to strictly enforce existing environmental policies is unlikely due to limited regulatory capacity but instead largely driven by the lack of bottom-up pressure. A promising direction of future research is therefore trying to understand how to get more citizens to spontaneously participate in environmental governance.

Finally, there is an extensive debate about the degree to which governments that are not held accountable by voting are accountable to their citizens. This paper's findings demonstrate that these governments still face important sources of accountability, and indeed the success of China's "War on Pollution" illustrates this point more broadly (Greenstone et al. 2021).

REFERENCES

- Acemoglu, Daron, Tarek A. Hassan, and Ahmed Tahoun. 2018. "The Power of the Street: Evidence from Egypt's Arab Spring." *Review of Financial Studies* 31 (1): 1–42.
- Adhvaryu, Achyuta, Namrata Kala, and Anant Nyshadham. 2016. "Management and shocks to worker productivity: evidence from air pollution exposure in an Indian garment factory." Unpublished.
- Allcott, Hunt, Luca Braghieri, Sarah Eichmeyer, and Matthew Gentzkow. 2020. "The Welfare Effects of Social Media." *American Economic Review* 110 (3): 629–76.
- Anderson, Sarah E., Mark T. Buntaine, Mengdi Liu, and Bing Zhang. 2019. "Non-Governmental Monitoring of Local Governments Increases Compliance with Central Mandates: A National-Scale Field Experiment in China." *American Journal of Political Science* 63 (3): 626–43.
- Bai, Chong-En, Chang-Tai Hsieh, and Zheng Song. 2020. "Special Deals with Chinese Characteristics." In *NBER Macroeconomics Annual*, Vol. 34, edited by Martin S. Eichenbaum, Erik Hurst, and Jonathan A. Parker, 341–79. Chicago, IL: University of Chicago Press.
- Banerjee, Abhijit V., Rukmini Banerji, Esther Duflo, Rachel Glennerster, and Stuti Khemani. 2012. "Pitfalls of Participatory Programs: Evidence from a Randomized Evaluation in Education in India." *American Economic Journal: Economic Policy* 2 (1): 1–30.
- Banerjee, Abhijit, and Esther Duflo. 2006. "Addressing Absence." *Journal of Economic Perspectives* 20 (1): 117–32.
- Bakshy, Eytan, Solomon Messing, and Lada A. Adamic. 2015. "Exposure to Ideologically Diverse News and Opinion on Facebook." *Science* 348 (6239): 1130–32.
- Beraja, Martin, Andrew Kao, David Y. Yang, and Noam Yuchtman. 2021. "AI-tocracy." NBER Working Paper 29466.
- Björkman, Martina, and Jakob Svensson. 2009. "Power to the People: Evidence from a Randomized Field Experiment on Community-Based Monitoring in Uganda." *Quarterly Journal of Economics* 124 (2): 735–69.

- Bui, Linda T. M., and Christopher J. Mayer.** 2003. "Regulation and Capitalization of Environmental Amenities: Evidence from the Toxic Release Inventory in Massachusetts." *Review of Economics and Statistics* 85 (3): 693–708.
- Buntaine, Mark, Michael Greenstone, Guojun He, Mengdi Liu, Shaoda Wang, and Bing Zhang.** 2024. "Data and code for: Does the Squeaky Wheel Get More Grease? The Direct and Indirect Effects of Citizen Participation on Environmental Governance in China." American Economic Association [publisher], Inter-university Consortium for Political and Social Research [distributor]. <https://doi.org/10.3886/E194521V1>.
- Buntaine, Mark T., Patrick Hunnicutt, and Polycarp Komakech.** 2021. "The Challenges of Using Citizen Reporting to Improve Public Services: A Field Experiment on Solid Waste Services in Uganda." *Journal of Public Administration Research and Theory* 31 (1): 108–27.
- Buntaine, Mark T., Bing Zhang, and Patrick Hunnicutt.** 2021. "Citizen Monitoring of Waterways Decreases Pollution in China by Supporting Government Action and Oversight." *Proceeding of the National Academy of Sciences* 18 (29): e2015175118.
- Burgess, Robin, Matthew Hansen, Benjamin A. Olken, Peter Potapov, and Stefanie Sieber.** 2012. "The Political Economy of Deforestation in the Tropics." *Quarterly Journal of Economics* 127 (4): 1707–54.
- Bursztyn, Leonardo, Georgy Egorov, Ruben Enikolopov, and Maria Petrova.** 2019. "Social Media and Xenophobia: Evidence from Russia." NBER Working Paper 26567.
- Campante, Filipe R., Davin Chor, and Bingjing Li.** 2019. "The Political Economy Consequences of China's Export Slowdown." NBER Working Paper 25925.
- Chang, Tom, Joshua Graff Zivin, Tal Gross, and Matthew Neidell.** 2016. "Particulate Pollution and the Productivity of Pear Packers." *American Economic Journal: Economic Policy* 8 (3): 141–69.
- Chen, Yuyu, and David Y. Yang.** 2019. "The Impact of Media Censorship: 1984 or Brave New World?" *American Economic Review* 109 (6): 2294–2332.
- Chen, Xi.** 2011. *Social Protest and Contentious Authoritarianism in China*. Cambridge: Cambridge University Press.
- Chen, Jidong, Jennifer Pan, and Yiqing Xu.** 2016. "Sources of Authoritarian Responsiveness: A Field Experiment in China." *American Journal of Political Science* 60 (2): 383–400.
- Crépon, Bruno, Esther Duflo, Marc Gurgand, Roland Rathelot, and Philippe Zamora.** 2013. "Do Labor Market Policies Have Displacement Effects? Evidence from a Clustered Randomized Experiment." *Quarterly Journal of Economics* 128 (2): 531–80.
- Currie, Janet, Lucas Davis, Michael Greenstone, and Reed Walker.** 2015. "Environmental Health Risks and Housing Values: Evidence from 1,600 Toxic Plant Openings and Closings." *American Economic Review* 105 (2): 678–709.
- Distelhorst, Greg, and Yue Hou.** 2017. "Constituency Service under Nondemocratic Rule: Evidence from China." *Journal of Politics* 79 (3): 1024–40.
- Duflo, Esther, Michael Greenstone, Rohini Pande, and Nicholas Ryan.** 2013. "Truth-Telling by Third-Party Auditors and the Response of Polluting Firms: Experimental Evidence from India." *Quarterly Journal of Economics* 128 (4): 1499–1545.
- Duflo, Esther, Michael Greenstone, Rohini Pande, and Nicholas Ryan.** 2018. "The Value of Regulatory Discretion: Estimates from Environmental Inspections in India." *Econometrica* 86 (6): 2123–60.
- Ebenstein, Avraham, Maoyong Fan, Michael Greenstone, Guojun He, and Maigeng Zhou.** 2017. "New Evidence on the Impact of Sustained Exposure to Air Pollution on Life Expectancy from China's Huai River Policy." *Proceedings of the National Academy of Sciences* 114 (39): 10384–89.
- Ebenstein, Avraham, and Michael Greenstone.** 2022. "Childhood Exposure to Particulate Air Pollution, Human Capital Accumulation, and Income: Evidence from China." Unpublished.
- Egger, Dennis, Johannes Haushofer, Edward Miguel, Paul Niehaus, and Michael W. Walker.** 2019. "General Equilibrium Effects of Cash Transfers: Experimental Evidence from Kenya." NBER Working Paper 26600.
- Enikolopov, Ruben, Alexey Makarin, and Maria Petrova.** 2020. "Social Media and Protest Participation: Evidence from Russia." *Econometrica* 88 (4): 1479–1514.
- European Union.** 2021. Monitoring, reporting and verification of EU ETS emissions.
- Garbiras-Díaz, Natalia, and Mateo Montenegro.** 2022. "All Eyes on Them: A Field Experiment on Citizen Oversight and Electoral Integrity." *American Economic Review* 112 (8): 2631–68.
- Gavazza, Alessandro, and Alessandro Lizzeri.** 2007. "The Perils of Transparency in Bureaucracies." *American Economic Review* 97 (2): 300–305.
- Gowrisankaran, Gautam, Michael Greenstone, Ali Hortaçsu, Mengdi Liu, Caixia Shen, and Bing Zhang.** 2020. "Pollution Mitigation and Productivity: Evidence from Chinese Manufacturing Firms." Unpublished.

- Graff Zivin, Joshua, and Matthew Neidell.** 2012. "The Impact of Pollution on Worker Productivity." *American Economic Review* 102 (7): 3652–73.
- Greenstone, Michael.** 2002. "The Impacts of Environmental Regulations on Industrial Activity: Evidence from the 1970 and 1977 Clean Air Act Amendments and the Census of Manufactures." *Journal of Political Economy* 110 (6): 1175–1219.
- Greenstone, Michael, and Rema Hanna.** 2014. "Environmental Regulations, Air and Water Pollution, and Infant Mortality in India." *American Economic Review* 104 (10): 3038–72.
- Greenstone, Michael, Guojun He, Shanjun Li, and Eric Yongchen Zou.** 2021. "China's War on Pollution: Evidence from the First 5 Years." *Review of Environmental Economics and Policy* 15 (2): 281–99.
- Greenstone, Michael, Guojun He, Ruixue Jia, and Tong Liu.** 2022. "Can Technology Solve the Principal-Agent Problem? Evidence from China's War on Air Pollution." *American Economic Review: Insights* 4 (1): 54–70.
- Greenstone, Michael, and B. Kelsey Jack.** 2015. "Envirodevonomics: A Research Agenda for an Emerging Field." *Journal of Economic Literature* 53 (1): 5–42.
- Greenstone, Michael, John A. List, and Chad Syverson.** 2012. "The Effects of Environmental Regulation on the Competitiveness of US Manufacturing." NBER Working Paper 18392.
- Greenstone, Michael, Janhavi Nilekani, Rohini Pande, Nicholas Ryan, Anant Sudarshan, and Anish Sugathan.** 2015. "Lower Pollution, Longer Lives: Life Expectancy Gains if India Reduced Particulate Matter Pollution." *Economic and Political Weekly* 50 (8): 40–46.
- Greenstone, Michael, Rohini Pande, Nicholas Ryan, and Anant Sudarshan.** 2020. *Continuous Emissions Monitoring Systems (CEMS) in India*. New Delhi: International Initiative for Impact Evaluation (3ie).
- Grossman, Guy, Melina R. Platas, and Jonathan Rodden.** 2018. "Crowdsourcing Accountability: ICT for Service Delivery." *World Development* 112: 74–87.
- Guo, Gang.** 2009. "China's Local Political Budget Cycles." *American Journal of Political Science* 53 (3): 621–32.
- He, Guojun, Shaoda Wang, and Bing Zhang.** 2020. "Watering Down Environmental Regulation in China." *Quarterly Journal of Economics* 135 (4): 2135–85.
- Henderson, J. Vernon.** 1996. "Effects of Air Quality Regulation." *American Economic Review* 86 (4): 789–813.
- Isen, Adam, Maya Rossin-Slater, and W. Reed Walker.** 2017. "Every Breath You Take—Every Dollar You'll Make: The Long-Term Consequences of the Clean Air Act of 1970." *Journal of Political Economy* 125 (3): 848–902.
- Ito, Koichiro, and Shuang Zhang.** 2020. "Willingness to Pay for Clean Air: Evidence from Air Purifier Markets in China." *Journal of Political Economy* 128 (5): 1627–72.
- Jia, Ruixue.** 2017. "Pollution for Promotion." Unpublished.
- Jing, Jun.** 2000. "Environmental Protests in Rural China." In *Chinese Society: Change, Conflict, and Resistance*, edited by Elizabeth J. Perry and Mark Selden, 60–143. New York, NY: Routledge.
- Kahn, Matthew E., Pei Li, and Daxuan Zhao.** 2015. "Water Pollution Progress at Borders: The Role of Changes in China's Political Promotion Incentives." *American Economic Journal: Economic Policy* 7 (4): 223–42.
- Kahn, Matthew E., and Erin T. Mansur.** 2013. "Do Local Energy Prices and Regulation Affect the Geographic Concentration of Employment?" *Journal of Public Economics* 101: 105–14.
- Konar, Shameek, and Mark A. Cohen.** 1997. "Information as Regulation: The Effect of Community Right to Know Laws on Toxic Emissions." *Journal of Environmental Economics and Management* 32 (1): 109–24.
- Kremer, Michael, Jessica Leino, Edward Miguel, and Alix Peterson Zwane.** 2011. "Spring Cleaning: Rural Water Impacts, Valuation, and Property Rights Institutions." *Quarterly Journal of Economics* 126 (1): 145–205.
- Lipscomb, Molly, and Ahmed Mushfiq Mobarak.** 2016. "Decentralization and Pollution Spillovers: Evidence from the Re-Drawing of County Borders in Brazil." *Review of Economic Studies* 84 (1): 464–502.
- List, John A., and Daniel M. Sturm.** 2006. "How Elections Matter: Theory and Evidence from Environmental Policy." *Quarterly Journal of Economics* 121 (4): 1249–81.
- Liu, Ernest, Yi Lu, Wenwei Peng, and Shaoda Wang.** 2022. "Judicial Independence, Local Protectionism, and Economic Integration: Evidence from China." NBER Working Paper 30432.
- Lorentzen, Peter L.** 2013. "Regularizing Rioting: Permitting Public Protest in an Authoritarian Regime." *Quarterly Journal of Political Science* 8 (2): 127–58.
- Mansuri, Ghazala, and Vijayendra Rao.** 2004. "Community-Based and-Driven Development: A Critical Review." *World Bank Research Observer* 19 (1): 1–39.

- Mattozzi, Andrea, and Antonio Merlo.** 2007. "The Transparency of Politics and the Quality of Politicians." *American Economic Review* 97 (2): 311–15.
- Mei, Yixin, and Yanhui Wu.** 2022. "Social Media and Government Responsiveness: Evidence from Vaccine Procurement in China." Unpublished.
- Olken, Benjamin A.** 2007. "Monitoring Corruption: Evidence from a Field Experiment in Indonesia." *Journal of Political Economy* 115 (2): 200–249.
- Qian, Yingyi, and Barry R. Weingast.** 1997. "Federalism as a Commitment to Reserving Market Incentives." *Journal of Economics Perspectives* 11 (4): 83–92.
- Qin, Bei, David Strömberg, and Yanhui Wu.** 2021. "Social Media and Collective Action in China." Unpublished.
- Reinikka, Ritva, and Jakob Svensson.** 2011. "The Power of Information in Public Services: Evidence from Education in Uganda." *Journal of Public Economics* 95 (7-8): 956–66.
- Ryan, Stephen P.** 2012. "The Costs of Environmental Regulation in a Concentrated Industry." *Econometrica* 80 (3): 1019–61.
- Steinert-Threlkeld, Zachary C.** 2017. "Spontaneous Collective Action: Peripheral Mobilization during the Arab Spring." *American Political Science Review* 111 (2): 379–403.
- Steinhardt, H. Christoph, and Fengshi Wu.** 2016. "In the Name of the Public: Environmental Protest and the Changing Landscape of Popular Contention in China." *China Journal* 75 (1): 61–82.
- Stiglitz, Joseph E.** 2002. "Participation and Development: Perspectives from the Comprehensive Development Paradigm." *Review of Development Economics* 6 (2): 163–82.
- United Nations Environment Program.** 2019. *Environmental Rule of Law: First Global Report*. Nairobi, Kenya: UN Environment.
- US EPA.** 2021. Enforcement Annual Results for Fiscal Year 2021.
- Walker, W. Reed.** 2013. "The Transitional Costs of Sectoral Reallocation: Evidence from the Clean Air Act and the Workforce." *Quarterly Journal of Economics* 128 (4): 1787–1835.
- Wang, Shaoda.** 2016. "Fiscal Competition and Coordination: Evidence from China." Unpublished.
- Wang, Shaoda, and David Y. Yang.** 2021. "Policy Experimentation in China: The Political Economy of Policy Learning." NBER Working Paper 29402.
- Wang, Shaoda, and Zenan Wang.** 2020. "The Environmental and Economic Consequences of Internalizing Border Spillovers." Unpublished.
- World Bank.** 2004. *World Development Report 2004: Making Services Work for Poor People*. Washington, DC: World Bank.
- Wu, Wenqi, Wenwen Wang, Lu Zhang, Qiaoyu Wang, Lujing Wang, and Ming Zhang.** 2021. "Does the Public Haze Pollution Concern Expressed on online Platforms Promoted Pollution Control? – Evidence from Chinese Online Platforms." *Journal of Cleaner Production* 318: 128477.
- Xu, Chenggang.** 2011. "The Fundamental Institutions of China's Reforms and Development." *Journal of Economic Literature* 49 (4): 1076–1151.
- Yanagizawa-Drott, David, Maria Petrova, and Ruben Enikolopov.** 2021. "Echo Chambers: Does Online Network Structure Affect Political Polarization?" Unpublished.
- Zhuravskaya, Ekaterina, Maria Petrova, and Ruben Enikolopov.** 2020. "Political Effects of the Internet and Social Media." *Annual Review of Economics* 12: 415–38.